

# Package ‘strategize’

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**Type** Package

**Title** Tools for Learning Adversarial or Non-Adversarial Optimal Distributions in High-Dimensional Conjoint Experiments

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**Description** The 'strategize' package implements methods for learning an optimal or adversarial probability distribution over high-dimensional factors in conjoint (or factorial) experiments. It supports both single-agent (non-adversarial) optimization and adversarial two-player settings, such as multi-stage electoral contexts. Users can estimate the distribution of factor levels that best achieves a chosen objective, optionally under institutional constraints (e.g., two-stage primaries). The package offers a variety of estimation routines, including closed-form solutions for some linear models, gradient-based optimizers for more complex outcome models, and built-in support for inference (via the delta method). It is particularly suitable for exploring and comparing candidate strategies in forced-choice conjoint studies, but is general enough for broader use in high-dimensional policy learning.

**URL** <https://github.com/cjerzak/strategize-software>

**BugReports** <https://github.com/cjerzak/strategize-software/issues>

**Depends** R ( $\geq 4.1$ )

**License** GPL-3

**Encoding** UTF-8

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jsonlite,  
stats,  
compositions,  
FactorHet,  
ggplot2,  
glinternet,  
graphics,  
gridExtra,  
gtools,  
Matrix,  
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ps,  
reticulate,  
rlang,  
rapply,

sandwich,  
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|                                              |                                                                   |
|----------------------------------------------|-------------------------------------------------------------------|
| <code>adapt_conjoint_foundation_model</code> | <i>Adapt a pooled conjoint foundation model to a single study</i> |
|----------------------------------------------|-------------------------------------------------------------------|

---

## Description

Adapt a pooled conjoint foundation model to a single study

## Usage

```
adapt_conjoint_foundation_model(
  foundation_model,
  Y,
  W,
  X = NULL,
  mode = c("auto", "pairwise", "single"),
  pair_id = NULL,
  profile_order = NULL,
  experiment_id = "adaptation_target",
  experiment_description = NULL,
  names_list = NULL,
  p_list = NULL,
  respondent_id = NULL,
  respondent_task_id = NULL,
  competing_group_variable_candidate = NULL,
  competing_group_variable_respondent = NULL,
  likelihood = NULL,
  n_outcomes = NULL,
  canonical_factor_id = NULL,
  canonical_level_id = NULL,
  neural_mcmc_control = NULL,
  foundation_adaptation_control = NULL,
  conda_env = "strategize_env",
  conda_env_required = TRUE,
  cache_path = NULL,
  cache_overwrite = FALSE,
  cache_compress = TRUE
)
```

## Arguments

|                               |                                                   |
|-------------------------------|---------------------------------------------------|
| <code>foundation_model</code> | A fitted <code>conjoint_foundation_model</code> . |
| <code>Y</code>                | Outcome vector.                                   |
| <code>W</code>                | Factor matrix/data.frame.                         |

|                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>X</code>                                   | Optional numeric covariates.                                                                                                                                                                                                                                                                                                                                                                                                   |
| <code>mode</code>                                | "auto", "pairwise", or "single".                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>pair_id</code>                             | Optional pair identifiers.                                                                                                                                                                                                                                                                                                                                                                                                     |
| <code>profile_order</code>                       | Optional within-pair ordering.                                                                                                                                                                                                                                                                                                                                                                                                 |
| <code>experiment_id</code>                       | Optional experiment identifier for the adaptation study.                                                                                                                                                                                                                                                                                                                                                                       |
| <code>experiment_description</code>              | Optional text description for the adaptation study, used when text-aware experiment tokens are enabled.                                                                                                                                                                                                                                                                                                                        |
| <code>names_list</code>                          | Optional factor level names.                                                                                                                                                                                                                                                                                                                                                                                                   |
| <code>p_list</code>                              | Optional <code>p_list</code> .                                                                                                                                                                                                                                                                                                                                                                                                 |
| <code>respondent_id</code>                       | Optional respondent identifiers.                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>respondent_task_id</code>                  | Optional respondent-task identifiers.                                                                                                                                                                                                                                                                                                                                                                                          |
| <code>competing_group_variable_candidate</code>  | Optional candidate competition-group labels for pairwise FM adaptation.                                                                                                                                                                                                                                                                                                                                                        |
| <code>competing_group_variable_respondent</code> | Optional respondent competition-group labels for pairwise FM adaptation.                                                                                                                                                                                                                                                                                                                                                       |
| <code>likelihood</code>                          | Optional likelihood override.                                                                                                                                                                                                                                                                                                                                                                                                  |
| <code>n_outcomes</code>                          | Optional categorical outcome count.                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>canonical_factor_id</code>                 | Optional factor-level sharing ids.                                                                                                                                                                                                                                                                                                                                                                                             |
| <code>canonical_level_id</code>                  | Optional level-level sharing ids.                                                                                                                                                                                                                                                                                                                                                                                              |
| <code>neural_mcmc_control</code>                 | Optional list passed to the Bayesian neural backend.                                                                                                                                                                                                                                                                                                                                                                           |
| <code>foundation_adaptation_control</code>       | Optional list controlling adaptation. Supported keys are <code>strict_schema_match</code> , <code>allow_extra_covariates</code> , <code>use_text_semantics</code> , <code>text_embedding_fn</code> , <code>experiment_token_mode</code> , <code>factor_tokenization</code> , <code>max_factor_tokens</code> , <code>covariate_value_encoding</code> , <code>shared_projection</code> , and <code>max_covariate_tokens</code> . |
| <code>conda_env</code>                           | Conda env name for the neural backend.                                                                                                                                                                                                                                                                                                                                                                                         |
| <code>conda_env_required</code>                  | Require the conda env to exist.                                                                                                                                                                                                                                                                                                                                                                                                |
| <code>cache_path</code>                          | Optional predictor cache path.                                                                                                                                                                                                                                                                                                                                                                                                 |
| <code>cache_overwrite</code>                     | Logical; overwrite any existing cache at <code>cache_path</code> .                                                                                                                                                                                                                                                                                                                                                             |
| <code>cache_compress</code>                      | Compression passed to <code>saveRDS()</code> .                                                                                                                                                                                                                                                                                                                                                                                 |

## Details

Current semantic zero-overlap foundation models are adapted canonically with preference. `fm::adapt_conjoint_foundation_model` This helper is retained for legacy runtime compatibility and reuses the package's existing Bayesian neural outcome model rather than fitting a separate downstream architecture. Internally, this function:

1. normalizes the target study into the same local schema representation used by the neural outcome model,

2. finds the compatible foundation group matching (mode, likelihood, n\_outcomes),
3. builds warm-start values from the pooled foundation parameters, and
4. runs the current Bayesian neural fit with those warm starts.

Group matching is exact. Adaptation fails if the foundation object does not contain a compatible internal family for the requested target study.

The target-study data contract mirrors the per-experiment contract used by `fit_conjoint_foundation_model()`:

- `length(Y) == nrow(W)`;
- pairwise studies require `pair_id`;
- `X`, when supplied, must be numeric and row-aligned to `W`; raw numeric values remain unchanged during adaptation. Under the default FM covariate path, shared covariate spans are aligned by pooled `X` name, name tokens optionally receive rebuilt `X`-name text embeddings, and value tokens are rebuilt from standardized numeric values plus local adaptation-study distribution summaries. Absent covariates are masked structurally rather than represented with a separate presence embedding;
- categorical adaptation follows the same `n_outcomes` rule as pooled training.

Schema reuse is partial by design:

- matched factors and levels inherit warm-started embeddings from the foundation fit;
- unmatched local schema elements fall back to local initialization;
- if `foundation_adaptation_control$strict_schema_match = TRUE`, unmatched factors cause an error instead of falling back.

The `foundation_adaptation_control` list supports:

`strict_schema_match` Require every local factor to match a pooled foundation slot. Default `FALSE`.

`allow_extra_covariates` Allow local covariates that were not present during pooled training. Extra local covariate spans are appended after the shared pooled covariate vocabulary. Default `TRUE`.

`use_text_semantics` Reuse token-native text semantics during adaptation when the foundation object includes them, including factor-name, level-label, and pooled `X`-name token components. Default `TRUE`.

`text_embedding_fn` Optional embedding function used to rebuild token-native text information for the target study. When omitted, adaptation reuses the stored pooled text registry when available. Any supplied function must return the same embedding width as the pooled foundation text registry.

`experiment_token_mode` Experiment token mode for the adapted fit. Defaults to the pooled group's FM setting.

`factor_tokenization` Factor tokenizer used during adaptation. Defaults to the pooled group's FM setting.

`max_factor_tokens` Total factor-token budget used during adaptation. Defaults to the pooled group's FM setting.

`covariate_value_encoding` Covariate encoder family used during adaptation. Defaults to the pooled group's FM setting.

`shared_projection_value_encoder` Value-token generator used inside "shared\_projection" during adaptation. Defaults to the pooled group's FM setting.

`max_covariate_tokens` Total covariate-token budget used during adaptation. Defaults to the pooled group's FM setting.

**Value**

A strategic\_predictor.

**See Also**

`fit_conjoint_foundation_model()`, `build_backend()`

**Examples**

```
## Not run:
library(strategize)

build_backend(conda_env = "strategize_env")

text_embedding_fn <- function(x) {
  x <- as.character(x)
  cbind(nchar = nchar(x), has_space = as.numeric(grepl(" ", x)))
}

foundation_fit <- preference.fm::fit_conjoint_foundation_model(
  experiments = list(
    list(
      experiment_id = "study_a",
      experiment_description = "Source policy-message study",
      Y = c(1, 0, 0, 1),
      W = data.frame(
        price = c("Low", "Low", "High", "High"),
        message = c("Jobs", "Taxes", "Jobs", "Taxes")
      ),
      pair_id = c(1, 1, 2, 2),
      profile_order = c(1, 2, 1, 2),
      canonical_factor_id = c(price = "price", message = "message")
    )
  ),
  foundation_control = list(text_embedding_fn = text_embedding_fn)
)

adapted_fit <- preference.fm::adapt_conjoint_foundation_model(
  foundation_model = foundation_fit,
  Y = c(1, 0, 0, 1),
  W = data.frame(
    price = c("Low", "High", "Low", "High"),
    message = c("Jobs", "Jobs", "Taxes", "Taxes")
  ),
  mode = "pairwise",
  pair_id = c(1, 1, 2, 2),
  profile_order = c(1, 2, 1, 2),
  experiment_id = "target_study",
  experiment_description = "Target policy-message study",
  canonical_factor_id = c(price = "price", message = "message"),
  foundation_adaptation_control = list(text_embedding_fn = text_embedding_fn)
)

predict(
  adapted_fit,
  newdata = list(
```

```

    W = data.frame(
      price = c("Low", "High"),
      message = c("Jobs", "Taxes")
    ),
    pair_id = c(1, 1),
    profile_order = c(1, 2)
  )
)

## End(Not run)

```

---

|             |                                                         |
|-------------|---------------------------------------------------------|
| as_function | <i>Convert a fitted predictor to a scoring function</i> |
|-------------|---------------------------------------------------------|

---

### Description

Convert a fitted predictor to a scoring function

### Usage

```
as_function(fit)
```

### Arguments

`fit`                      A fitted strategic\_predictor.

### Value

A closure that calls `predict()` (or `predict_pair()` for pairwise).

---

|               |                                                           |
|---------------|-----------------------------------------------------------|
| build_backend | <i>Build the computational environment for strategize</i> |
|---------------|-----------------------------------------------------------|

---

### Description

Creates the conda environment used by the package's JAX-backed optimization workflow, neural APIs, and foundation checkpoint loading. Users may also create and manage a compatible environment themselves.

### Usage

```

build_backend(
  conda_env = "strategize_env",
  conda = "auto",
  backend = c("auto", "cpu", "cuda", "mps"),
  force_reinstall = FALSE
)

```

**Arguments**

|                 |                                                                                                                                                                                                                                                                                      |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| conda_env       | Name of the conda environment in which to place the backends. Defaults to "strategize_env".                                                                                                                                                                                          |
| conda           | The path to a conda executable. Using "auto" allows reticulate to attempt to automatically find an appropriate conda binary. Defaults to "auto". If creation fails and a conda binary is found on PATH, the function retries with it.                                                |
| backend         | JAX backend selector. Use "auto" to preserve the default host-aware behavior, "cpu" to install plain CPU JAX, "cuda" to require Linux and install CUDA-capable JAX wheels when supported by the NVIDIA driver, or "mps" to opt in to the experimental Apple Silicon jax-mps backend. |
| force_reinstall | Logical. If TRUE, remove the named conda environment if it exists, then recreate and reinstall the backend from the requested input specifications.                                                                                                                                  |

**Details**

The default backend = "auto" preserves the existing non-MPS behavior. The "mps" backend is experimental, opt-in, and supported only on macOS Apple Silicon. It creates a Python 3.13 environment, installs jax-mps, verifies that JAX reports the MPS backend under JAX\_PLATFORMS=mps, and recreates an existing incompatible environment.

**Value**

Invisibly returns NULL. This function is called for its side effects of creating and configuring a conda environment for strategize. This function requires an Internet connection. You can find a list of conda Python paths via: `Sys.which("python")`

**Examples**

```
## Not run:
# Create a conda environment named "strategize_env"
# and install the required Python packages (jax, numpy, orbox-checkpoint, etc.)
build_backend(conda_env = "strategize_env", conda = "auto")

# If you want to specify a particular conda path:
# build_backend(conda_env = "strategize_env", conda = "/usr/local/bin/conda")

# Experimental Apple Silicon MPS backend:
# build_backend(backend = "mps")
# Sys.setenv(JAX_PLATFORMS = "mps") may be needed before importing JAX
# directly through reticulate in an already-running R session.

## End(Not run)
```

---

build\_text\_embedding\_backend

*Inspect or build a host-aware text embedding backend*

---



## Description

Resolve a text embedding backend that is compatible with the current machine while preserving a portable Qwen-family embedding surface across Mac and Linux hosts.

## Usage

```
build_text_embedding_backend(
    conda_env = "strategize_env",
    family = "qwen3",
    runtime = "auto",
    profile = "portable",
    cache_dir = NULL,
    required = TRUE,
    conda = "auto"
)

inspect_text_embedding_backend(
    conda_env = "strategize_env",
    family = "qwen3",
    runtime = "auto",
    profile = "portable",
    conda = "auto"
)
```

## Arguments

|           |                                                                                                                                   |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------|
| conda_env | Conda env name used for the Python runtime.                                                                                       |
| family    | Embedding-family selector. Currently only "qwen3" is supported.                                                                   |
| runtime   | Runtime preference. Use "auto" to resolve per host, or choose one of "mlx", "cuda", "rocm", or "cpu" explicitly.                  |
| profile   | Embedding profile. Currently only "portable" is supported.                                                                        |
| cache_dir | Optional directory for cached text embeddings.                                                                                    |
| required  | Logical; if TRUE, ensure the selected runtime is usable immediately. If FALSE, return a lazy backend that validates on first use. |
| conda     | Conda binary to use. Defaults to "auto".                                                                                          |

## Details

The current implementation keeps the Qwen embedding family but selects a different runtime per host:

- Apple Silicon defaults to MLX with `mlx-community/Qwen3-Embedding-8B-mxfp8`.
- Linux prefers a CUDA-capable `sentence-transformers` path on validated Nvidia hosts.
- Linux prefers a ROCm-capable `sentence-transformers` path when the Python runtime validates it.
- Other hosts fall back to a CPU `sentence-transformers` path.

All runtimes are normalized to a canonical 1024-dimensional embedding width so saved artifacts remain portable across hosts.

**Value**

inspect\_text\_embedding\_backend() returns a serializable list describing the host, Python env health, candidate runtimes, and the selected backend.

build\_text\_embedding\_backend() returns a list with \$fn, \$spec, and \$runtime.

**Examples**

```
## Not run:
backend <- build_text_embedding_backend(conda_env = "strategize_env")

study_a$experiment_description <- "Source study A"
study_b$experiment_description <- "Source study B"

foundation_fit <- preference.fm::fit_conjoint_foundation_model(
  experiments = list(study_a, study_b),
  foundation_control = list(
    text_embedding_fn = backend$fn
  )
)

inspect_text_embedding_backend(conda_env = "strategize_env")

## End(Not run)
```

---

check\_hessian\_geometry

*Check Hessian Geometry at Nash Equilibrium*

---

**Description**

Computes Hessian eigenvalues to verify proper saddle point structure at the Nash equilibrium. For a valid Nash equilibrium in a zero-sum game:

- AST (maximizer) Hessian should be negative semi-definite (all eigenvalues  $\leq 0$ )
- DAG (minimizer) Hessian should be positive semi-definite (all eigenvalues  $\geq 0$ )

**Usage**

```
check_hessian_geometry(result, tol = 1e-06, verbose = TRUE)
```

**Arguments**

|         |                                                                |
|---------|----------------------------------------------------------------|
| result  | Output from <a href="#">strategize</a> with adversarial = TRUE |
| tol     | Numeric. Tolerance for near-zero eigenvalues (default 1e-6)    |
| verbose | Logical. Whether to print progress messages. Default is TRUE.  |

## Details

### Mathematical Foundation

At a Nash equilibrium in a zero-sum game, the Hessian matrices encode local curvature information:

- **AST (maximizes Q)**: The Hessian should be negative semi-definite, meaning all eigenvalues are  $\leq 0$ . This confirms AST is at a local maximum.
- **DAG (minimizes Q)**: The Hessian should be positive semi-definite, meaning all eigenvalues are  $\geq 0$ . This confirms DAG is at a local minimum.

The condition number (ratio of largest to smallest eigenvalue magnitude) indicates equilibrium robustness. High condition numbers suggest the equilibrium is sensitive to small perturbations.

### Interpretation

- `status = "PASS"`: Valid saddle point with no flat directions
- `status = "WARNING"`: Valid saddle point but has flat directions (weak identification). Consider increasing regularization.
- `status = "FAIL"`: Not a proper saddle point. The solution may not have converged to a true Nash equilibrium.

## Value

A list of class "hessian\_analysis" containing:

**status** Character: "PASS", "WARNING", or "FAIL"

**valid\_saddle** Logical. TRUE if both Hessians have correct definiteness

**eigenvalues\_ast** Eigenvalues of AST player's Hessian

**eigenvalues\_dag** Eigenvalues of DAG player's Hessian

**is\_negative\_semidefinite\_ast** Logical. TRUE if AST Hessian is negative semi-definite

**is\_positive\_semidefinite\_dag** Logical. TRUE if DAG Hessian is positive semi-definite

**condition\_number\_ast** Condition number of AST Hessian (stability indicator)

**condition\_number\_dag** Condition number of DAG Hessian

**flat\_directions\_ast** Number of near-zero eigenvalues (weak identification)

**flat\_directions\_dag** Number of near-zero eigenvalues

**interpretation** Character string with human-readable interpretation

Returns NULL with a warning if Hessian functions are not available.

## See Also

[validate\\_equilibrium](#) for best-response validation

## Examples

```
## Not run:
# Run adversarial strategize
result <- strategize(Y = y, W = w, adversarial = TRUE, nSGD = 500)

# Check Hessian geometry
hess <- check_hessian_geometry(result)
print(hess)
```

```
# Check if it's a valid saddle point
if (hess$valid_saddle) {
  message("Valid Nash equilibrium geometry!")
}

## End(Not run)
```

---

conjoint-foundation      *Conjoint Foundation Models*

---

## Description

Load pooled conjoint foundation-model artifacts produced by **preference.fm**.

## Details

Foundation-model work now follows a split workflow:

1. Train pooled foundation models with `preference.fm::fit_conjoint_foundation_model()`.
2. Save them as checkpoint directories with `preference.fm::save_conjoint_foundation_bundle()`.
3. Load bundles, extract embeddings, and run predictors with **strategize**; adapt semantic foundation models with `preference.fm::adapt_conjoint_foundation_model()`.

Pooled training does not force every experiment into one universal output head. Experiments are grouped into compatible neural families defined by (mode, likelihood, n\_outcomes). Each compatible family shares one pooled schema-aware encoder and one neural fit, while incompatible families are stored as separate internal groups in the returned `conjoint_foundation_model`.

Cross-study sharing is driven by explicit canonical ids. Raw label equality alone does not force two studies to share factor or level identities. Optional text embeddings now enter the FM encoder directly: factor spans use language-native factor-name and level-label tokens, and covariate spans use covariate-name tokens plus value tokens conditioned on local covariate distributions. Canonical ids still determine schema sharing across studies.

`fit_conjoint_foundation_model()` and `save_conjoint_foundation_bundle()` are retained as migration stubs that point to **preference.fm**. Use `preference.fm::adapt_conjoint_foundation_model()` for semantic target-study adaptation.

---

country\_latlong\_cache      *Country latitude/longitude cache*

---

## Description

Return the bundled country lookup table used by foundation-model place context tokens. The cache contains ISO codes, country names and aliases, and representative latitude/longitude values.

## Usage

```
country_latlong_cache()
```

**Value**

A data frame with one row per country.

---

|               |                                                   |
|---------------|---------------------------------------------------|
| create_p_list | <i>Create Baseline Probability List from Data</i> |
|---------------|---------------------------------------------------|

---

**Description**

Generates a p\_list suitable for the strategize() function from observed data or using uniform probabilities.

**Usage**

```
create_p_list(W, uniform = FALSE)
```

**Arguments**

|         |                                                                                                                        |
|---------|------------------------------------------------------------------------------------------------------------------------|
| W       | Factor matrix (data.frame or matrix) with one column per conjoint factor.                                              |
| uniform | Logical; if TRUE, use uniform probabilities across levels; if FALSE (default), use observed frequencies from the data. |

**Details**

For conjoint experiments with balanced randomization, use uniform = TRUE. For experiments with intentional imbalance or to match observed frequencies, use uniform = FALSE.

**Value**

Named list where each element is a named numeric vector of probabilities. Names correspond to factor levels.

**Examples**

```
# Create sample factor matrix
W <- data.frame(
  Gender = c("Male", "Female", "Male", "Female"),
  Age = c("Young", "Old", "Young", "Old")
)

# Uniform probabilities (for balanced designs)
p_list_uniform <- create_p_list(W, uniform = TRUE)
print(p_list_uniform)
# $Gender
#   Male Female
#   0.5    0.5
# $Age
#   Young   Old
#   0.5    0.5

# Observed frequencies
p_list_observed <- create_p_list(W, uniform = FALSE)
print(p_list_observed)
```

---

|               |                                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| cv_strategize | <i>Cross-validation for Optimal Stochastic Interventions in Conjoint Analysis</i> |
|---------------|-----------------------------------------------------------------------------------|

---

## Description

Performs cross-validation to select the regularization parameter  $\lambda$  (and, if desired, other hyperparameters) for the `strategize` function. This function splits the data by respondent (or user-specified units), trains candidate models under a grid of  $\lambda$  values, and evaluates out-of-sample performance, returning the model that maximizes a chosen criterion (e.g., out-of-sample expected utility or log-likelihood).

## Usage

```
cv_strategize(
  Y,
  W,
  X = NULL,
  lambda_seq = NULL,
  lambda = NULL,
  folds = 2L,
  crossfit_q = FALSE,
  crossfit_q_control = NULL,
  varcov_cluster_variable = NULL,
  competing_group_variable_respondent = NULL,
  competing_group_variable_candidate = NULL,
  competing_group_competition_variable_candidate = NULL,
  pair_id = NULL,
  respondent_id = NULL,
  respondent_task_id = NULL,
  profile_order = NULL,
  p_list = NULL,
  slate_list = NULL,
  use_optax = F,
  K = 1,
  nSGD = 100,
  diff = F,
  adversarial = F,
  adversarial_model_strategy = "four",
  partial_pooling = NULL,
  partial_pooling_strength = 50,
  use_regularization = TRUE,
  force_gaussian = F,
  force_reinforce = FALSE,
  temperature = NULL,
  a_init_sd = 0.001,
  learning_rate_max = 0.001,
  penalty_type = "KL",
  outcome_model_type = "glm",
  neural_mcmc_control = NULL,
  compute_se = T,
```

```

conda_env = "strategize_env",
conda_env_required = F,
conf_level = 0.9,
nFolds_glm = 3L,
nMonte_adversarial = 5L,
primary_pushforward = "mc",
primary_strength = 1,
primary_n_entrants = 1L,
primary_n_field = 1L,
nMonte_Qglm = 100L,
optim_type = "gd",
optimism = "extragrad",
optimism_coef = 1,
rain_lambda = 1,
rain_gamma = 0.01,
rain_L = NULL,
rain_eta = 0.001,
rain_variant = "alg10_staged",
rain_output = "last"
)

```

## Arguments

|                         |                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Y                       | A numeric or binary response vector. If binary (e.g., 0–1), it should correspond to forced-choice outcomes (1 if candidate A is chosen; 0 if candidate B is chosen). If numeric, please see details in <a href="#">strategize</a> for how outcomes are handled.                                                                                                                          |
| W                       | A data frame or matrix representing the randomized conjoint attributes. Each column is a factor or character vector indicating attribute levels for a particular dimension. Multiple columns can be used if the conjoint has multiple attributes.                                                                                                                                        |
| X                       | Optional covariate matrix or data frame for modeling systematic heterogeneity. If $K > 1$ , this is typically required for multi-class or cluster-based models. Otherwise, set $X = \text{NULL}$ .                                                                                                                                                                                       |
| lambda_seq              | A numeric vector of candidate $\lambda$ values for cross-validation. If $\text{NULL}$ and $\lambda$ is also $\text{NULL}$ , a sequence of values is automatically generated (e.g., via $10^{\text{seq}(-4, 0, \text{length.out} = 5)} * \text{sd}(Y)$ ).                                                                                                                                 |
| lambda                  | A single user-specified $\lambda$ value. If provided, cross-validation is effectively disabled unless $\lambda_{\text{seq}}$ is also supplied.                                                                                                                                                                                                                                           |
| folds                   | An integer or user-specified partitioning indicating the number of cross-validation folds. Defaults to 2. See Details for how data splitting is done.                                                                                                                                                                                                                                    |
| crossfit_q              | Logical. If $\text{TRUE}$ , compute $Q_{\text{crossfit}}$ , $Q_{\text{reference\_crossfit}}$ , $Q_{\text{gain\_crossfit}}$ , and $Q_{\text{crossfit\_info}}$ on the final refit after $\lambda$ selection. Supported for non-adversarial pairwise average-case binomial GLM runs and adversarial pairwise binomial GLM runs with $\text{adversarial\_model\_strategy} = \text{"four"}$ . |
| crossfit_q_control      | Optional list passed to <a href="#">strategize</a> to control cross-fitted $Q$ evaluation, including the optional adversarial perspective_group entry.                                                                                                                                                                                                                                   |
| varcov_cluster_variable | An optional clustering variable for robust standard errors. For instance, if the data is from multiple respondents, specify respondent IDs here for cluster-robust inference (via sandwich estimation). If $\text{NULL}$ , no cluster-based variance correction is used.                                                                                                                 |

|                                                |                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| competing_group_variable_respondent            | Optional vector for multi-round or multi-group setups, indicating which respondent belongs to which group. Used for advanced or adversarial designs (e.g., dual-party contexts). If NULL, standard usage is assumed.                                                                                                                                                                                                         |
| competing_group_variable_candidate             | Similar to competing_group_variable_respondent, but for candidate-level grouping. If NULL, standard usage is assumed.                                                                                                                                                                                                                                                                                                        |
| competing_group_competition_variable_candidate | An optional variable for specifying which candidate is in competition with which group. Relevant if multi-step adversarial frameworks are used.                                                                                                                                                                                                                                                                              |
| pair_id                                        | An optional vector (same length as Y) identifying which rows (candidate pairs) belong to the same forced choice. For example, if each respondent evaluates multiple pairs, this ID ensures correct grouping. Required only in certain advanced difference-in-differences or paired analyses.                                                                                                                                 |
| respondent_id                                  | A user-specified ID to denote respondent-level grouping, typically used to cluster standard errors or to perform out-of-sample validation by respondent. If NULL, a simple row index is used for splitting.                                                                                                                                                                                                                  |
| respondent_task_id                             | Another optional ID for tasks (e.g., each respondent might see multiple tasks). Helps in advanced designs. If NULL, ignored.                                                                                                                                                                                                                                                                                                 |
| profile_order                                  | An optional vector capturing the ordering of candidate profiles within tasks, if multiple profiles are being shown. Used in difference or extended hierarchical modeling.                                                                                                                                                                                                                                                    |
| p_list                                         | A list of assignment probabilities for each attribute, if known or desired as a baseline. If NULL, each level is assumed to have uniform probability or derived from empirical frequencies in W.                                                                                                                                                                                                                             |
| slate_list                                     | An optional list specifying alternative or restricted sets of attribute levels. Used when a subset of attributes is feasible or when bounding certain strategies in an adversarial design.                                                                                                                                                                                                                                   |
| use_optax                                      | Logical. If TRUE, uses the <b>optax</b> Python library (via <b>reticulate</b> ) for gradient-based optimization. If FALSE, uses a default gradient-based approach from <b>jax</b> .                                                                                                                                                                                                                                          |
| K                                              | An integer specifying the number of mixture components or clusters if X is used (e.g., for multi-class analysis). Defaults to 1 (no mixture).                                                                                                                                                                                                                                                                                |
| nSGD                                           | An integer number of iterations for gradient-based training. Defaults to 100 but can be increased if convergence has not been reached.                                                                                                                                                                                                                                                                                       |
| diff                                           | Logical indicating whether a difference-based model (e.g., for forced-choice or difference-in-outcomes) is used. Defaults to FALSE, but set TRUE in certain difference-of-utility designs.                                                                                                                                                                                                                                   |
| adversarial                                    | Logical indicating whether to use a two-party or multi-agent <i>adversarial</i> approach in the optimization. If TRUE, a min-max (zero-sum) formulation is employed. Defaults to FALSE (single-agent or average-case optimization).                                                                                                                                                                                          |
| adversarial_model_strategy                     | Character string indicating whether to estimate "four" outcome models (primary + general for each group), "two" outcome models (one per group reused for both primary and general), or "neural" (Bayesian Transformer models with party tokens; defaults to a single pooled model across groups and stages. Set <code>neural_mcmc_control\$n_bayesian_models = 2</code> to fit separate AST/DAG models) in adversarial mode. |



**partial\_pooling**

Logical indicating whether to partially pool (shrink) group-specific outcome model coefficients toward a shared average when using the "two" strategy. When NULL, defaults to TRUE in the two-strategy adversarial case.

**partial\_pooling\_strength**

Numeric scalar controlling the amount of shrinkage used for partial pooling in the two-strategy adversarial case.

**use\_regularization**

Logical; if TRUE, penalty-based regularization is used for the outcome model. Usually set to TRUE for large designs. Defaults to TRUE.

**force\_gaussian**

Logical indicating whether a Gaussian family (lm-style) is forced for the outcome model, even if Y is binary. Defaults to FALSE.

**force\_reinforce**

Logical indicating whether to force REINFORCE-based optimization for the neural average-case objective when exact small-support enumeration is available. This affects the optimization objective only; final reported Q values still use the standard report-time evaluation path. Defaults to FALSE.

**temperature**

Optional numeric temperature controlling the smoothness of Gumbel-Softmax sampling when exploring probability vectors. Smaller values lead to distributions closer to the argmax. Defaults to NULL, which allows internal defaults.

**a\_init\_sd**

A numeric controlling the random initialization scale for unconstrained parameters in gradient-based optimization. Defaults to 0.001. Larger values can help avoid local minima in complex outcome landscapes.

**learning\_rate\_max**

Base learning rate for gradient-based optimizers. Defaults to 0.001.

**penalty\_type**

A character string specifying the type of penalty for the *optimal stochastic intervention*, e.g., "KL", "L2", or "LogMaxProb". The default is "KL".

**outcome\_model\_type**

Character string indicating the outcome model to use, such as "glm" for generalized linear models or "neural" for a neural-network approximation. Defaults to "glm".

**neural\_mcmc\_control**

Optional list overriding default MCMC settings used when outcome\_model\_type = "neural". Use neural\_mcmc\_control\$uncertainty\_scope = "output" to restrict delta-method uncertainty to output-layer parameters. In adversarial neural mode, set neural\_mcmc\_control\$n\_bayesian\_models = 2 to fit separate AST/DAG models (default is 1 for a single differential model). Use neural\_mcmc\_control\$ModelDim and neural\_mcmc\_control\$ModelDepth to override the Transformer hidden width and depth. Pairwise neural fits default to neural\_mcmc\_control\$cross\_candidate\_encoder = "term" when unspecified, except rank-positive low-rank respondent-candidate interaction defaults to "none" to avoid duplicating pairwise interaction capacity. Set neural\_mcmc\_control\$cross\_candidate\_encoder = "term" (or TRUE) to include the opponent-dependent cross-candidate term in pairwise mode, set neural\_mcmc\_control\$cross\_candidate\_encoder = "attn" to add a lightweight cross-attention step. When combined with neural\_mcmc\_control\$cross\_candidate\_encoder = "full\_attn", that step consumes the depth-attended candidate-token readout. Set neural\_mcmc\_control\$cross\_candidate\_encoder = "full" to enable a full cross-encoder that jointly encodes both candidates. Use "none" (or FALSE) to disable. Set neural\_mcmc\_control\$residual\_mode = "full\_attn" to replace the default additive/ReZero residual path with full depth-wise attention over all prior layer outputs plus a final depth-attentive readout; use "standard"

(default) to keep the existing residual formulation. Rank-positive low-rank Bernoulli pairwise logits use RMS logit normalization by default; set `neural_mcmc_control$low_rank_logit_transform = "none"` to disable it. Explicit `low_rank_logit_transform = "softclip"` remains supported for compatibility. For variational inference (`subsample_method = "batch_vi"`), set `neural_mcmc_control$optimizer` to `"muon"` (default when `optax.contrib.muon` is available), `"adam"` (`numpyro.optim`), `"adamw"` (AdamW), or `"adabelief"` (`optax`). Muon is only applied to matrix-valued model-weight locations when the guide preserves matrix structure; with `vi_guide = "auto_diagonal"`, it falls back to AdamW/Adam. Learning-rate decay is controlled by `neural_mcmc_control$svi_steps` (integer steps, or `"optimal"` for a scaling-law heuristic based on model/data size; for minibatched VI this also scales with `batch_size`) and `neural_mcmc_control$svi_lr_schedule` (default `"warmup_cosine"`), with optional `svi_lr_warmup_frac` and `svi_lr_end_factor`. Validation-based early stopping is enabled by default for SVI-backed neural fits; set `neural_mcmc_control$early_stopping = FALSE` to force the full `svi_steps` budget. Use `neural_mcmc_control$early_stopping_n_checks` (default 10) to request an approximate number of validation checks during SVI early stopping; the cadence is derived as `ceiling(svi_steps / early_stopping_n_checks)`. Use `neural_mcmc_control$early_stopping_patience` (default 3) to control how many consecutive non-improving validation checks are tolerated before stopping. For compact streaming SVI, `early_stopping = TRUE` enables validation best-checkpoint selection but still runs the full `svi_steps` budget. Set `neural_mcmc_control$early_stopping_validation_frac` (default 0.05) to retain approximately that fraction of evaluable observations in the held-out validation split, and optionally `neural_mcmc_control$early_stopping_validation_max_n` (default 2048) to cap the retained validation size after fraction-based sizing. Set `early_stopping_validation_max_n = NULL` to disable that cap. Checkpoint gradient diagnostics are enabled by default for SVI fits and are evaluated at the early-stopping checkpoint cadence; set `neural_mcmc_control$gradient_diagnostics = FALSE` to disable them.

|                                  |                                                                                                                                                                                                                                                                                                  |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>compute_se</code>          | Logical; if TRUE, attempts to compute standard errors using M-estimation or the Delta method. Defaults to TRUE.                                                                                                                                                                                  |
| <code>conda_env</code>           | A character specifying the name of a Conda environment for the JAX-backed optimization workflow. Defaults to <code>"strategize_env"</code> .                                                                                                                                                     |
| <code>conda_env_required</code>  | Logical. If TRUE, errors if the specified Conda environment <code>conda_env</code> cannot be found. Otherwise tries to fall back gracefully.                                                                                                                                                     |
| <code>conf_level</code>          | The confidence level (between 0 and 1) for interval estimation, default 0.90.                                                                                                                                                                                                                    |
| <code>nFolds_glm</code>          | An integer specifying the number of folds in internal regression-based cross-validation (if used) for outcome model selection. Defaults to 3.                                                                                                                                                    |
| <code>nMonte_adversarial</code>  | A positive integer specifying the number of Monte Carlo draws for the min-max (adversarial) stage, if <code>adversarial = TRUE</code> . Defaults to 5.                                                                                                                                           |
| <code>primary_pushforward</code> | Character string controlling the primary-stage push-forward estimator. Use <code>"mc"</code> (default) for Monte Carlo sampling with per-draw primary winners, or <code>"linearized"</code> for the faster averaged-weight approximation, or <code>"multi"</code> for multi-candidate primaries. |
| <code>primary_strength</code>    | Numeric scalar controlling primary decisiveness (see <a href="#">strategize</a> ).                                                                                                                                                                                                               |
| <code>primary_n_entrants</code>  | Integer number of entrant candidates per party in multi-candidate primaries.                                                                                                                                                                                                                     |

|                 |                                                                                                                                                                                                                                                                                                                                            |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| primary_n_field | Integer number of field candidates per party in multi-candidate primaries.                                                                                                                                                                                                                                                                 |
| nMonte_Qglm     | An integer specifying the number of Monte Carlo draws for evaluating certain integrals in glm-based approximations, default 100.                                                                                                                                                                                                           |
| optim_type      | A character describing the optimization routine. Typically "default" uses a standard gradient-based approach; set "tryboth" or "SecondOrder" for testing or advanced routines.                                                                                                                                                             |
| optimism        | Character string controlling optimistic / extra-gradient updates for the gradient optimizer. Options: "extragrad" (default), "smp" (stochastic mirror-prox), "ogda", "rain" (RAIN: Recursive Anchored Iteration with anchored extra-gradient and increasing quadratic anchor penalties), or "none". Only supported when use_optax = FALSE. |
| optimism_coef   | Numeric scalar controlling the magnitude of optimism adjustments. For "ogda", this scales the optimistic correction term. Ignored when optimism = "rain".                                                                                                                                                                                  |
| rain_lambda     | Numeric scalar giving the base RAIN regularization scale $\lambda$ . The staged algorithm uses $\lambda_0 = \gamma\lambda$ and grows by $(1+\gamma)$ each stage. Only used when optimism = "rain".                                                                                                                                         |
| rain_gamma      | Non-negative numeric scalar for the RAIN anchor-growth parameter $\gamma$ . If not supplied, the default is auto-scaled downward when nSGD exceeds 100 to keep total anchor growth roughly constant. Only used when optimism = "rain".                                                                                                     |
| rain_L          | Optional numeric scalar for the Lipschitz constant $L$ used by RAIN. When supplied and rain_eta is missing, rain_eta defaults to $1/(8L)$ and stage lengths follow $T_s = 16L/\lambda_s$ . If NULL, $L$ is estimated from rain_eta. Only used when optimism = "rain".                                                                      |
| rain_eta        | Optional numeric scalar step size $\eta$ for RAIN. Defaults to 0.001 and is auto-scaled downward when nSGD exceeds 100 if not supplied. If rain_L is supplied and rain_eta is missing, defaults to $1/(8L)$ . Only used when optimism = "rain".                                                                                            |
| rain_variant    | Character string specifying the RAIN variant. Options: "alg10_staged" (merged Algorithm 10 with recursive stage anchors; default) or "alg9_single_loop" (single-loop variant; not yet implemented). Only used when optimism = "rain".                                                                                                      |
| rain_output     | Character string controlling the stage output for RAIN. "uniform_half" samples uniformly from half-iterates within the stage (most faithful); "last" returns the last iterate (default). Only used when optimism = "rain".                                                                                                                 |

## Details

cv\_strategize implements a cross-validation routine for [strategize](#). First, the data is split into folds parts. For each fold, we train candidate outcome models and compute out-of-sample performance. The best-performing  $\lambda$  is selected. Finally, a refit on the full data is done using the chosen hyperparameters, returning the results of the final [strategize](#) call with  $\lambda$  set to the best value.

The function supports a wide range of conjoints, including forced-choice (where diff = TRUE), multi-cluster outcome modeling (where  $K > 1$ ), and adversarial designs (where adversarial = TRUE). Regularization for the outcome model or for the candidate distribution can be enabled via use\_regularization and penalty\_type. Cross-validation is particularly helpful when the data is limited or highly dimensional.

## Value

A named list with components:

- pi\_star\_point** The estimated optimal probability distribution(s) over candidate profiles ( $\hat{\pi}^*$ ).
- Q\_point** The primary estimated expected outcome (e.g., vote share) under the selected optimal distribution.
- Q\_se** The primary standard error for **Q\_point**, when standard errors are requested.
- Q\_point\_mEst, Q\_se\_mEst** Backward-compatible aliases for **Q\_point** and **Q\_se**.
- lambda** The chosen  $\lambda$  value from cross-validation (and any other relevant hyperparameters).
- CVInfo** A data frame or matrix summarizing cross-validation results, e.g., in-sample and out-of-sample estimates for each candidate  $\lambda$ .
- Q\_crossfit, Q\_reference\_crossfit, Q\_gain\_crossfit, Q\_crossfit\_info** Optional final-refit cross-fitted policy-value diagnostics when `crossfit_q = TRUE`.
- Other components** Various additional objects useful for inference and debugging (e.g., final model fits, standard error estimates, weighting details).

### See Also

[strategize](#) for direct optimization of stochastic interventions in conjoint analysis, including average and adversarial settings.

### Examples

```
# =====
# Cross-validation to select regularization lambda
# =====
set.seed(123)
n <- 400 # profiles (200 pairs)

# Generate factor matrix
W <- data.frame(
  Gender = sample(c("Male", "Female"), n, replace = TRUE),
  Age = sample(c("35", "50", "65"), n, replace = TRUE),
  Party = sample(c("Dem", "Rep"), n, replace = TRUE)
)

# Simulate outcome with true effects
latent <- 0.2 * (W$Gender == "Female") + 0.15 * (W$Age == "35")
prob <- plogis(latent)

# Create paired forced-choice structure
pair_id <- rep(1:(n/2), each = 2)
Y <- numeric(n)
for (p in unique(pair_id)) {
  idx <- which(pair_id == p)
  winner <- sample(idx, 1, prob = prob[idx])
  Y[idx] <- as.integer(seq_along(idx) == which(idx == winner))
}
profile_order <- rep(1:2, n/2)

# Cross-validate over lambda values
# Lower lambda = less regularization = further from baseline
cv_result <- cv_strategize(
  Y = Y,
  W = W,
  lambda_seq = c(0.01, 0.1, 0.5, 1.0),
  folds = 2,
```

```

    pair_id = pair_id,
    respondent_id = pair_id,
    profile_order = profile_order,
    diff = TRUE,
    nSGD = 50,
    compute_se = FALSE
  )

  # View CV results and selected lambda
  print(cv_result$lambda)      # Optimal lambda
  print(cv_result$CVInfo)     # Performance at each lambda
  print(cv_result$pi_star_point) # Optimal distribution
  print(cv_result$Q_point)    # Expected outcome

```

---

|                    |                                                                 |
|--------------------|-----------------------------------------------------------------|
| extract_embeddings | <i>Extract last-layer embeddings from fitted neural objects</i> |
|--------------------|-----------------------------------------------------------------|

---

## Description

Extract the final hidden representation immediately before the neural output head, or the model's respondent-context representation.

## Usage

```

extract_embeddings(object, ...)

## S3 method for class 'strategic_predictor'
extract_embeddings(
  object,
  newdata,
  level = c("candidate", "respondent_context", "all"),
  ...
)

## S3 method for class 'conjoint_foundation_model'
extract_embeddings(
  object,
  newdata,
  group_key = NULL,
  mode = c("auto", "pairwise", "single"),
  likelihood = NULL,
  n_outcomes = NULL,
  level = c("candidate", "respondent_context", "all"),
  experiment_id = NULL,
  names_list = NULL,
  p_list = NULL,
  canonical_factor_id = NULL,
  canonical_level_id = NULL,
  foundation_adaptation_control = NULL,
  conda_env = "strategize_env",

```

```

conda_env_required = TRUE,
...
)

```

## Arguments

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| object                        | A fitted neural object. Supported classes are <code>strategic_predictor</code> and <code>conjoint_foundation_model</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| ...                           | Additional method-specific arguments.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| newdata                       | New data. For <code>strategic_predictor</code> , this follows the same contract as <code>predict()</code> . For <code>conjoint_foundation_model</code> , use either a list containing <code>W</code> and optional <code>X</code> , <code>pair_id</code> , <code>profile_order</code> , and <code>competing_group_variable_candidate</code> , <code>competing_group_variable_respondent</code> , and <code>experiment_id</code> , or a data frame/matrix. When supplying a raw data frame to the foundation method, extra numeric <code>X</code> columns can only be separated from factor columns when <code>names_list</code> (or <code>p_list</code> ) identifies the factor columns. |
| group_key                     | Optional internal foundation group key to extract from.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| mode                          | Optional mode filter used for foundation-group selection.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| likelihood                    | Optional likelihood filter used for foundation-group selection.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| n_outcomes                    | Optional outcome-count filter used for foundation-group selection.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| level                         | Embedding level to return. "candidate" preserves the historical candidate/profile output, "respondent_context" returns one respondent-context matrix, and "all" returns both.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| experiment_id                 | Optional experiment identifier used for experiment-token lookup.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| names_list                    | Optional local factor-level metadata for raw foundation extraction.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| p_list                        | Optional <code>p_list</code> used to derive <code>names_list</code> when needed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| canonical_factor_id           | Optional canonical factor ids used to map local factors into a pooled foundation group.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| canonical_level_id            | Optional canonical level ids used to map local levels into a pooled foundation group.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| foundation_adaptation_control | Optional list controlling schema matching for raw foundation extraction. Supported keys are <code>strict_schema_match</code> and <code>allow_extra_covariates</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| conda_env                     | Conda env name for neural extraction when params must be reconstructed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| conda_env_required            | Require the conda env to exist.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

## Details

`strategic_predictor` extraction works on fitted neural predictors, including objects returned by `strategic_prediction()` and `adapt_conjoint_foundation_model()`.

`conjoint_foundation_model` extraction runs directly on a selected pooled foundation group. When extracting from a raw foundation object, local studies must be mappable into the pooled schema, typically via matching `experiment_id`, canonical ids, or direct use of pooled slot names and level keys.

Respondent-context embeddings are derived from the model's context-side tokens: respondent group, respondent covariates, experiment tokens, and pairwise stage/matchup tokens when those components are available.

This function does not extend `predict()` or `predict_pair()`; it is a dedicated embedding API.

**Value**

An object of class `strategic_embeddings`.

For `level = "candidate"`, single-mode models contain `$embeddings`, an  $n \times d$  numeric matrix.

For pairwise models:

- non-"full" cross-candidate encoders return `$left` and `$right` matrices with one row per pair;
- "full" cross-candidate encoders return `$joint`, the pair-level final readout used immediately before the output head.

For `level = "respondent_context"`, the return value contains `$respondent_context`, an  $n \times d$  or pair-level  $n_{pairs} \times d$  matrix. For `level = "all"`, the candidate fields and `$respondent_context` are returned together.

All outputs also include `$metadata`, including the source class, mode, requested level, cross-candidate encoder mode, embedding width, and context component availability.

**Examples**

```
## Not run:
fit <- strategic_prediction(
  Y = y,
  W = w,
  model = "neural",
  mode = "single"
)

emb <- extract_embeddings(fit, newdata = w[1:10, , drop = FALSE])

foundation_emb <- extract_embeddings(
  foundation_fit,
  newdata = list(
    W = target_w,
    X = target_x,
    pair_id = target_pair_id,
    profile_order = target_profile_order
  ),
  group_key = "pairwise::bernoulli::1",
  canonical_factor_id = c(price = "price", message = "message")
)

## End(Not run)
```

---

`fit_conjoint_foundation_model`

*Pooled conjoint foundation-model training moved to preference.fm*

---

**Description**

Pooled conjoint foundation-model training moved to preference.fm

## Usage

```
fit_conjoint_foundation_model(
  experiments,
  foundation_control = NULL,
  conda_env = "strategize_env",
  conda_env_required = TRUE,
  cache_path = NULL,
  cache_overwrite = FALSE,
  cache_compress = TRUE
)
```

## Arguments

**experiments** List of experiment specifications. Each element must be a named list with at least `experiment_id`, `Y`, and `W`.

**foundation\_control** Optional list controlling pooled training. Supported keys are `add_experiment_indicators`, `add_text_semantics`, `text_embedding_fn`, and `neural_mcmc_control`.

**conda\_env** Conda env name for the neural backend.

**conda\_env\_required** Require the conda env to exist.

**cache\_path** Optional path to a cached foundation bundle.

**cache\_overwrite** Logical; overwrite any existing cache at `cache_path`.

**cache\_compress** Retained for API compatibility.

## Details

This exported name is retained for migration guidance. Pooled training is now owned by **preference.fm**. Use `preference.fm::fit_conjoint_foundation_model()` to train pooled models, `preference.fm::adapt_conjoint_foundation_model()` for semantic adaptation, and **strategize** for loading, embedding extraction, and prediction.

Each element of `experiments` is a per-study specification with the following contract.

Required fields:

**experiment\_id** A unique study identifier.

**Y** Outcome vector with `length(Y) == nrow(W)`.

**W** A data frame or matrix of factor columns.

Optional fields:

**mode** "auto", "pairwise", or "single". Defaults to "pairwise" when `pair_id` is supplied and "single" otherwise.

**pair\_id** Required for pairwise studies. Each pair id must appear exactly twice.

**profile\_order** Optional within-pair ordering, typically 1/2.

**X** Optional numeric covariates aligned row-wise to `W`. Non-numeric columns are rejected. Raw numeric values remain unchanged at the API boundary. In the default FM covariate path (`covariate_value_encoding = "shared_projection"` with `shared_projection_value_encoder = "name_dist_moe"`), each present covariate is emitted as a 4-token span `<start_covariate,`



covariate\_name\_token, value\_token, end\_covariate>. The name token can include projected X-name text embeddings, and the value token combines the standardized numeric value with local study-specific distribution summaries and covariate-name semantics. Absent covariates are masked structurally rather than represented by a separate learned presence embedding.

respondent\_id, respondent\_task\_id Optional row-aligned respondent/task identifiers used when available for clustering and evaluation logic.

competing\_group\_variable\_candidate, competing\_group\_variable\_respondent Optional row-aligned competition-group labels for pairwise studies. When both are supplied and the candidate labels span both same-group and cross-group pairs, the pooled FM trains in a stage-aware pairwise mode. Otherwise the pairwise FM remains stage-free.

likelihood "auto", "bernoulli", "categorical", or "normal".

n\_outcomes Required only when forcing a categorical likelihood. In v1 it must match the number of observed classes in the study. Categorical outcomes are internally normalized to zero-based class ids before neural fitting.

names\_list, p\_list Optional factor-level metadata used to define the local level universe for each factor.

canonical\_factor\_id Optional named vector or list that forces cross-study sharing of factor identities when explicitly provided.

canonical\_level\_id Optional named list that forces cross-study sharing of level identities when explicitly provided.

Pooled training groups experiments by backend-compatible task mode. Pairwise context mode is a supported capability of a pairwise group rather than a split key: stage-free and stage-aware pairwise studies can share one group. If any pooled pairwise study requires stage-aware context, the shared pairwise group is promoted to stage-aware-capable and records all supported pairwise context modes. The returned object may therefore hold multiple internal foundation groups when the input studies mix pairwise and single designs.

Schema sharing rules are conservative:

- explicit canonical ids force sharing;
- absent factors in a pooled schema are routed to holdout rows during training;
- raw text equality alone does not merge schema elements.

The foundation\_control list supports:

add\_experiment\_indicators Whether to append experiment indicators. This field is retained for compatibility, but the foundation neural path now always represents experiment identity as an explicit experiment token rather than one-hot covariates.

add\_text\_semantics Whether to add token-native text semantics when text\_embedding\_fn is supplied. This covers projected factor-name embeddings, level-label embeddings, and pooled X-name embeddings inside the emitted tokens. Default TRUE.

text\_embedding\_fn Optional function that maps character input to numeric embeddings. It may accept a character vector and return a matrix with matching rows, or accept one string at a time and return a fixed-width numeric vector. These embeddings are projected into FM factor-name, factor-level, and pooled X-name token components. Canonical ids still govern cross-study pooling and schema sharing.

experiment\_token\_mode Experiment token mode for FM training. "description" uses experiment-description text, "hybrid" combines description text and learned pooled experiment ids, and "legacy\_id" uses only learned pooled experiment ids. Default "description".

**factor\_tokenization** FM factor tokenizer. The default "language\_span" emits factor spans as `<start_factor, factor_name_token, factor_level_token, end_factor>`. "legacy\_indexed" retains the older fixed one-token-per-factor encoder for ablations. Default "language\_span".

**max\_factor\_tokens** Total factor-token budget used by the FM factor span encoder. The default 256L supports up to 64 factor spans per profile.

**covariate\_value\_encoding** FM covariate encoder family. "shared\_projection" is the span-based default. "legacy\_linear" retains the older fixed-width covariate path for ablations. Default "shared\_projection".

**shared\_projection\_value\_encoder** Value-token generator used inside "shared\_projection". The default "name\_dist\_moe" conditions each covariate value token on covariate-name semantics and local distribution summaries. "legacy\_scalar" falls back to the older standardized scalar projection.

**max\_covariate\_tokens** Total covariate-token budget used by the FM covariate span encoder. The default 512L supports up to 128 covariate spans per row.

**neural\_mcmc\_control** Optional list passed to the existing neural outcome backend. Defaults to a pooled SVI configuration using output-only uncertainty.

## Value

An object of class `conjoint_foundation_model`.

## See Also

[adapt\\_conjoint\\_foundation\\_model\(\)](#), [save\\_conjoint\\_foundation\\_bundle\(\)](#), [load\\_conjoint\\_foundation\\_bundle\(\)](#), [build\\_backend\(\)](#)

## Examples

```
## Not run:
library(strategize)

build_backend(conda_env = "strategize_env")

text_embedding_fn <- function(x) {
  x <- as.character(x)
  cbind(nchar = nchar(x), has_space = as.numeric(grepl(" ", x)))
}

study_a <- list(
  experiment_id = "study_a",
  experiment_description = "Pairwise policy-message study",
  Y = c(1, 0, 0, 1),
  W = data.frame(
    price = c("Low", "Low", "High", "High"),
    message = c("Jobs", "Taxes", "Jobs", "Taxes")
  ),
  pair_id = c(1, 1, 2, 2),
  profile_order = c(1, 2, 1, 2),
  canonical_factor_id = c(price = "price", message = "message")
)

study_b <- list(
  experiment_id = "study_b",
  experiment_description = "Pairwise messenger study",
```

```

Y = c(0, 1, 1, 0),
W = data.frame(
  price = c("Low", "High", "Low", "High"),
  message = c("Jobs", "Jobs", "Taxes", "Taxes"),
  messenger = c("Local", "Local", "National", "National")
),
pair_id = c(1, 1, 2, 2),
profile_order = c(1, 2, 1, 2),
canonical_factor_id = c(
  price = "price",
  message = "message",
  messenger = "messenger"
)
)
)

foundation_fit <- preference.fm::fit_conjoint_foundation_model(
  experiments = list(study_a, study_b),
  foundation_control = list(
    text_embedding_fn = text_embedding_fn,
    neural_mcmc_control = list(
      ModelDims = 32L,
      ModelDepth = 1L,
      subsample_method = "batch_vi",
      uncertainty_scope = "output",
      svi_steps = 100L
    )
  )
)
)

## End(Not run)

```

---

|                     |                                 |
|---------------------|---------------------------------|
| get_final_gradients | <i>Get Final Gradient Norms</i> |
|---------------------|---------------------------------|

---

## Description

Extract the final gradient magnitudes from a strategize result.

## Usage

```
get_final_gradients(result)
```

## Arguments

result                      Output from [strategize](#)

## Value

Named numeric vector with gradient norms for AST and DAG players.

---

|         |                                                    |
|---------|----------------------------------------------------|
| helpers | <i>User-Facing Helper Functions for strategize</i> |
|---------|----------------------------------------------------|

---

**Description**

Convenience functions to simplify common tasks when using the strategize package. These functions reduce the complexity of the main API by providing sensible defaults and automatic data processing.

---

|                                |                                                    |
|--------------------------------|----------------------------------------------------|
| inspect_text_embedding_backend | <i>Inspect a host-aware text-embedding backend</i> |
|--------------------------------|----------------------------------------------------|

---

**Description**

Inspect a host-aware text-embedding backend

**Usage**

```
inspect_text_embedding_backend(  
    conda_env = "strategize_env",  
    family = "qwen3",  
    runtime = "auto",  
    profile = "portable",  
    conda = "auto"  
)
```

**Arguments**

|           |                                                                                                                  |
|-----------|------------------------------------------------------------------------------------------------------------------|
| conda_env | Conda env name used for the Python runtime.                                                                      |
| family    | Embedding-family selector. Currently only "qwen3" is supported.                                                  |
| runtime   | Runtime preference. Use "auto" to resolve per host, or choose one of "mlx", "cuda", "rocm", or "cpu" explicitly. |
| profile   | Embedding profile. Currently only "portable" is supported.                                                       |
| conda     | Conda binary to use. Defaults to "auto".                                                                         |

**Value**

A serializable list describing the host, Python env health, candidate text-embedding runtimes, and the selected backend.

---

|                |                                       |
|----------------|---------------------------------------|
| is_adversarial | <i>Check if Result is Adversarial</i> |
|----------------|---------------------------------------|

---

### Description

Utility function to check if a strategize result is from adversarial mode.

### Usage

```
is_adversarial(result)
```

### Arguments

|        |                                        |
|--------|----------------------------------------|
| result | Output from <a href="#">strategize</a> |
|--------|----------------------------------------|

### Value

Logical. TRUE if the result is from adversarial mode.

---

|                                 |                                          |
|---------------------------------|------------------------------------------|
| load_conjoint_foundation_bundle | <i>Load a conjoint foundation bundle</i> |
|---------------------------------|------------------------------------------|

---

### Description

Load a conjoint foundation bundle

### Usage

```
load_conjoint_foundation_bundle(
  file,
  conda_env = "strategize_env",
  conda_env_required = TRUE,
  preload_params = FALSE
)
```

### Arguments

|                    |                                                                                                                                                  |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| file               | Path to a checkpoint directory created by <code>preference.fm::save_conjoint_foundation_bundle()</code> or to a legacy <code>.rds</code> bundle. |
| conda_env          | Conda env name for the neural backend.                                                                                                           |
| conda_env_required | Require the conda env to exist.                                                                                                                  |
| preload_params     | Logical; reconstruct neural params immediately.                                                                                                  |

### Details

Loading reconstructs a `conjoint_foundation_model` without importing or depending on **preference.fm**. New checkpoint directories restore direct neural parameter PyTrees from Orbax. Legacy `.rds` foundation bundles remain readable for older artifacts.

**Value**

A conjoint\_foundation\_model.

**See Also**

[save\\_conjoint\\_foundation\\_bundle\(\)](#)

**Examples**

```
## Not run:  
foundation_fit <- load_conjoint_foundation_bundle("foundation_bundle")  
  
## End(Not run)
```

---

load\_neural\_outcome\_bundle

*Load a portable neural outcome bundle*

---

**Description**

Load a portable neural outcome bundle

**Usage**

```
load_neural_outcome_bundle(  
  file,  
  conda_env = "strategize_env",  
  conda_env_required = TRUE,  
  preload_params = FALSE  
)
```

**Arguments**

|                    |                                                                         |
|--------------------|-------------------------------------------------------------------------|
| file               | Path to a bundle created by <code>save_neural_outcome_bundle()</code> . |
| conda_env          | Conda env name for neural backend. Use NULL to defer to metadata.       |
| conda_env_required | Require conda env to exist for neural backend.                          |
| preload_params     | Logical; if TRUE, reconstruct neural params immediately.                |

**Value**

A strategic\_predictor ready for `predict()` / `predict_pair()`.

---

load\_strategic\_predictor

*Load a strategic predictor from disk*


---

### Description

Load a strategic predictor from disk

### Usage

```
load_strategic_predictor(
  file,
  conda_env = "strategize_env",
  conda_env_required = TRUE
)
```

### Arguments

|                    |                                                                                 |
|--------------------|---------------------------------------------------------------------------------|
| file               | Path to a cached predictor created by save_strategic_predictor().               |
| conda_env          | Conda env name for neural predictors. Use NULL to defer to the cached metadata. |
| conda_env_required | Require conda env to exist for neural predictors.                               |

### Value

A strategic\_predictor ready for predict().

---

plot\_best\_response\_curves

*Plot Dimension-by-Dimension Best-Response Curves from Adversarial strategize() Output*


---

### Description

plot\_best\_response\_curves takes the result of an adversarial [strategize](#) run (i.e., with adversarial = TRUE) and produces dimension-specific best-response curves. Specifically, for a chosen factor dimension  $d$ , it plots:

1. The curve of  $\pi_{\text{dag},d}^*$  as a function of  $\pi_{\text{ast},d}$ .
2. The curve of  $\pi_{\text{ast},d}^*$  as a function of  $\pi_{\text{dag},d}$ .

Potential intersection points in this 2D space can indicate approximate equilibria for dimension  $d$ , holding the other dimensions fixed at the solution found by strategize.

This function is computationally intensive: for each of nPoints\_br grid values of  $\pi_{\text{ast},d}$ , it searches over possible  $\pi_{\text{dag},d}$  (and vice versa) to find each side's best response, re-running partial objective evaluations. Nonetheless, it provides a direct visualization of how each player (ast or dag) responds to changes in the other's distribution along a single factor dimension.

## Usage

```
plot_best_response_curves(
  res,
  d_ = 1,
  nPoints_br = 100L,
  nPoints_heat = 50L,
  title = NULL,
  col_ast = "blue",
  col_dag = "red",
  lwd_ast = 2,
  lwd_dag = 2,
  point_pch = 19,
  silent = FALSE
)
```

## Arguments

|                  |                                                                                                                                                                                                                                                                                                                                                             |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| res              | A list returned by <code>strategize</code> , which must include adversarial references. Internally, res should contain items like <code>res\$a_i_ast</code> , <code>res\$a_i_dag</code> , the JAX-based functions ( <code>dQ_da_ast</code> , <code>dQ_da_dag</code> , <code>QFXN</code> , ...), along with the appropriate unconstrained parameter vectors. |
| d_               | (Integer) The dimension of $\pi_{ast}, \pi_{dag}$ to examine. For example, if you have multiple factors (dimensions), each is indexed by a positive integer. Defaults to 1.                                                                                                                                                                                 |
| nPoints_br       | (Integer) Number of equally spaced grid points in $[0, 1]$ to sample for <i>the outer</i> loop. The code does an internal small search for best responses at each grid point. Larger nPoints_br => smoother curves but more computation. Defaults to 100L.                                                                                                  |
| nPoints_heat     | (Integer) Number of grid points for heat map calculations. Defaults to 50L.                                                                                                                                                                                                                                                                                 |
| title            | (Character or NULL) Main plot title. If NULL, an auto-generated title is used, e.g. "Best-Response Curves (Dimension d_=1)".                                                                                                                                                                                                                                |
| col_ast, col_dag | (Character) Colors for ast's and dag's best-response curves, respectively. Defaults to "blue" (ast) and "red" (dag).                                                                                                                                                                                                                                        |
| lwd_ast, lwd_dag | (Numeric) Line widths for ast and dag curves, respectively. Default is 2.                                                                                                                                                                                                                                                                                   |
| point_pch        | (Numeric) Symbol for marking the approximate intersection (if found) on the plot. Defaults to 19 (filled circle).                                                                                                                                                                                                                                           |
| silent           | (Logical) If TRUE, suppresses printed messages during the search for intersection. Defaults to FALSE.                                                                                                                                                                                                                                                       |

## Details

**Mechanics:** For each  $\pi_{ast,d} \in \{0, \frac{1}{nPoints_{br}-1}, \dots, 1\}$ , the function temporarily fixes that dimension in the ast player's unconstrained parameter vector. It then does an internal grid search over  $\pi_{dag,d}$  to see which value yields the largest dag payoff (lowest ast payoff), consistent with `adversarial=TRUE`. The resulting curve is  $BR_{dag}(\pi_{ast,d})$ .

Likewise, it holds  $\pi_{dag,d}$  fixed and searches over  $\pi_{ast,d}$  to get  $BR_{ast}(\pi_{dag,d})$ .



The intersection in  $(\pi_{\text{ast},d}, \pi_{\text{dag},d})$ -space (if one exists in the discretized grid) is a candidate local equilibrium for that factor dimension, *given that the other factor dimensions remain at the solution from [strategize](#)*.

**Performance Caution:** This brute force line-search re-runs partial objective evaluations many times, which may be slow for large `nPoints_br` or complex outcome models. Consider using a smaller `nPoints_br` if performance is an issue, or focusing on only a handful of crucial dimensions  $d$ .

## Value

(Invisibly) A list containing:

`grid_points` A numeric vector of the `nPoints_br` grid values in  $[0, 1]$  used for the outer loop.

`br_dag_given_ast` A numeric vector of the same length, giving the dag best-response  $\pi_{\text{dag},d}$  at each grid point for  $\pi_{\text{ast},d}$ .

`br_ast_given_dag` A numeric vector with the ast best-response for each grid point  $\pi_{\text{dag},d}$ .

## See Also

[strategize](#) for obtaining the result object `res` in adversarial mode. See also [cv\\_strategize](#).

## Examples

```
## Not run:
# =====
# Visualize best-response curves in adversarial mode
# =====
# First, fit an adversarial strategize model
set.seed(42)
n <- 400

# Generate data with party structure
W <- data.frame(
  Gender = sample(c("Male", "Female"), n, replace = TRUE),
  Age = sample(c("Young", "Middle", "Old"), n, replace = TRUE)
)

# Party affiliations for respondents and candidates
respondent_party <- sample(c("Dem", "Rep"), n/2, replace = TRUE)
candidate_party <- rep(c("Dem", "Rep"), n/2)

Y <- rbinom(n, 1, 0.5) # Simplified outcome

# Fit adversarial model
adv_result <- strategize(
  Y = Y,
  W = W,
  lambda = 0.1,
  adversarial = TRUE,
  competing_group_variable_respondent = rep(respondent_party, each = 2),
  competing_group_variable_candidate = candidate_party,
  nSGD = 100
)

# Plot best-response curves for Gender dimension (d_ = 1)
```

```
# Shows how each party's optimal Gender distribution responds
# to changes in the other party's Gender distribution
plot_best_response_curves(
  res = adv_result,
  d_ = 1, # Gender is first factor
  nPoints_br = 50,
  title = "Gender: Best-Response Curves",
  col_ast = "blue", # Democrats
  col_dag = "red"   # Republicans
)

# Intersection point indicates Nash equilibrium for this dimension

## End(Not run)
```

plot\_convergence

*Plot Convergence Diagnostics for Strategize Results***Description**

Visualizes the optimization trajectory from gradient descent, showing gradient magnitudes, loss values, and learning rate adaptation over iterations.

**Usage**

```
plot_convergence(
  result,
  metrics = c("gradient", "loss"),
  log_scale = TRUE,
  use_ggplot = FALSE
)
```

**Arguments**

|            |                                                                                                                                                                                                                                                                                                                                              |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| result     | Output from <a href="#">strategize</a> with <code>adversarial = TRUE</code>                                                                                                                                                                                                                                                                  |
| metrics    | Character vector specifying which metrics to plot. Options are: <ul style="list-style-type: none"> <li>"gradient": Gradient magnitude (L2 norm) over iterations</li> <li>"loss": Objective function value over iterations</li> <li>"lr": Learning rate adaptation over iterations</li> </ul> Default is <code>c("gradient", "loss")</code> . |
| log_scale  | Logical. Whether to use log scale for gradient magnitudes. Default is TRUE.                                                                                                                                                                                                                                                                  |
| use_ggplot | Logical. If TRUE and ggplot2 is available, uses ggplot2 for visualization. Otherwise uses base R graphics. Default is FALSE.                                                                                                                                                                                                                 |

**Details**

This function provides diagnostic plots to assess whether the gradient descent optimization has converged to a Nash equilibrium. Key indicators of convergence:

- Gradient magnitudes should decrease toward zero for both players

- Loss values should stabilize (may oscillate slightly in adversarial settings)
- Learning rates should adapt appropriately (decrease as gradients shrink)

In adversarial (two-player) mode, both AST and DAG players' metrics are shown. In non-adversarial mode, only AST metrics are displayed.

### Value

If `use_ggplot = TRUE` and `ggplot2` is available, returns a `ggplot` object. Otherwise, plots are created as side effects and the function returns `invisible(NULL)`.

### Examples

```
## Not run:
# Run adversarial strategize
result <- strategize(Y = y, W = w, adversarial = TRUE, nSGD = 500)

# Plot convergence diagnostics
plot_convergence(result)

# Plot only gradient magnitudes with log scale
plot_convergence(result, metrics = "gradient", log_scale = TRUE)

## End(Not run)
```

---

plot\_quadrant\_breakdown

*Plot Four-Quadrant Contribution Breakdown*

---

### Description

Decomposes the equilibrium vote share  $Q^*$  into contributions from four distinct election scenarios, providing insight into which primary-to-general election pathways drive the Nash equilibrium outcome.

### Usage

```
plot_quadrant_breakdown(
  result,
  type = c("bar", "pie"),
  nMonte = 500,
  verbose = TRUE
)
```

### Arguments

|                     |                                                                                                                                                                                               |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>result</code> | Output from <code>strategize</code> with <code>adversarial = TRUE</code>                                                                                                                      |
| <code>type</code>   | Character string specifying the plot type: <ul style="list-style-type: none"> <li>• "bar": Stacked bar chart showing contributions</li> <li>• "pie": Pie chart showing proportions</li> </ul> |

|         |                                                                        |
|---------|------------------------------------------------------------------------|
|         | Default is "bar".                                                      |
| nMonte  | Integer. Number of Monte Carlo samples for estimation. Default is 500. |
| verbose | Logical. Whether to print progress messages. Default is TRUE.          |

## Details

In adversarial mode, two parties simultaneously optimize their candidate attribute distributions. Each party's "entrant" candidate (drawn from the optimized distribution) competes in a primary against a "field" candidate (drawn from the baseline distribution). The general election outcome depends on which candidates win their respective primaries, creating four possible scenarios. This function visualizes the relative importance of each scenario to the overall equilibrium.

The four scenarios (quadrants) represent different primary election outcomes:

**E1: Both Entrants** Both parties' "entrant" candidates (sampled from optimized distributions) win their primaries

**E2: A Entrant, B Field** Party A's entrant wins, Party B's field candidate (sampled from baseline) wins

**E3: A Field, B Entrant** Party A's field wins, Party B's entrant wins

**E4: Both Field** Both parties' field candidates win their primaries

The weight of each scenario depends on the primary election probabilities (kappa values), which in turn depend on the voter model.

## Value

A list containing:

**weights** Named vector of four-quadrant weights (sum to 1)

**contributions** Named vector of contributions to  $Q^*$  from each quadrant

**Q\_star** Total equilibrium vote share

**plot** Base R plot (invisible return)

## Interpretation

- A dominant E1 contribution suggests entrant vs. entrant matchups are most important for the equilibrium
- Balanced contributions indicate a robust equilibrium across scenarios
- Large E4 contribution suggests field candidates often win primaries

## Examples

```
## Not run:
# Run adversarial strategize
result <- strategize(Y = y, W = w, adversarial = TRUE, nSGD = 500)

# Plot quadrant breakdown
breakdown <- plot_quadrant_breakdown(result)
print(breakdown$weights)
print(breakdown$contributions)

## End(Not run)
```

---

```
predict.strategic_predictor
```

*Predict from a fitted strategic predictor*

---

## Description

Predict from a fitted strategic predictor

## Usage

```
## S3 method for class 'strategic_predictor'
predict(
  object,
  newdata,
  type = c("response", "link"),
  interval = c("none", "ci", "draws"),
  level = 0.95,
  n_draws = 0L,
  seed = NULL,
  ...
)
```

## Arguments

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| object   | A fitted strategic_predictor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| newdata  | <p>New data. For mode="single", a data.frame/matrix of factor columns. For mode="pairwise", either:</p> <ul style="list-style-type: none"> <li>a data.frame containing factor columns plus pair_id (and optionally profile_order), or</li> <li>a list with elements W, pair_id, and optional profile_order.</li> </ul> <p>Neural foundation predictors also accept optional experiment_country and experiment_year as scalar or row-aligned vectors; in pairwise mode row-aligned values are taken from the left/profile-level prediction row.</p> |
| type     | "response" (probability) or "link" (logit / linear predictor).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| interval | "none" (default), "ci", or "draws".                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| level    | Credible interval level for draws.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| n_draws  | Number of posterior draws when interval!="none".                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| seed     | Optional seed for draws.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| ...      | Unused.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

---

```
prediction-api
```

*Prediction-Only Outcome Model API*

---

## Description

Fit the package's outcome model (GLM or neural) without running the stochastic intervention optimization in [strategize](#). Returns a fitted strategic\_predictor object with a [predict](#) method.

---

|              |                                             |
|--------------|---------------------------------------------|
| predict_pair | <i>Predict on wide-format pairwise data</i> |
|--------------|---------------------------------------------|

---

## Description

Predict on wide-format pairwise data

## Usage

```
predict_pair(
  fit,
  W_left,
  W_right,
  type = c("response", "link"),
  interval = c("none", "ci", "draws"),
  level = 0.95,
  n_draws = 0L,
  seed = NULL,
  ...
)
```

## Arguments

|          |                                                                                                                                                                                                                    |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| fit      | A fitted stage-free strategic_predictor with mode="pairwise". Stage-aware pairwise predictors require long-format <a href="#">predict()</a> input because candidate and respondent group metadata are row-aligned. |
| W_left   | Data frame/matrix of left profiles.                                                                                                                                                                                |
| W_right  | Data frame/matrix of right profiles.                                                                                                                                                                               |
| type     | "response" or "link".                                                                                                                                                                                              |
| interval | "none", "ci", or "draws".                                                                                                                                                                                          |
| level    | Credible interval level for draws.                                                                                                                                                                                 |
| n_draws  | Number of posterior draws when interval!="none".                                                                                                                                                                   |
| seed     | Optional seed for draws.                                                                                                                                                                                           |
| ...      | Unused.                                                                                                                                                                                                            |

## Value

Predictions for each row-pair.

---

```
print.hessian_analysis
```

*Print method for hessian\_analysis objects*

---

### Description

Print method for hessian\_analysis objects

### Usage

```
## S3 method for class 'hessian_analysis'  
print(x, ...)
```

### Arguments

|     |                                                                       |
|-----|-----------------------------------------------------------------------|
| x   | A hessian_analysis object from <a href="#">check_hessian_geometry</a> |
| ... | Additional arguments (ignored)                                        |

---

```
print.strategize_result
```

*Print Method for strategize Results*

---

### Description

Print Method for strategize Results

### Usage

```
## S3 method for class 'strategize_result'  
print(x, digits = 3, ...)
```

### Arguments

|        |                                |
|--------|--------------------------------|
| x      | A strategize result object     |
| digits | Number of digits to display    |
| ...    | Additional arguments (ignored) |

### Value

Invisibly returns the input object

---

```
save_conjoint_foundation_bundle
```

*Conjoint foundation bundle writing moved to preference.fm*

---

## Description

Conjoint foundation bundle writing moved to preference.fm

## Usage

```
save_conjoint_foundation_bundle(
  file,
  foundation_model,
  overwrite = FALSE,
  compress = TRUE
)
```

## Arguments

|                  |                                       |
|------------------|---------------------------------------|
| file             | Path to save the bundle.              |
| foundation_model | A fitted conjoint_foundation_model.   |
| overwrite        | Logical; overwrite any existing file. |
| compress         | Retained for API compatibility.       |

## Details

This exported name is retained for migration guidance. New foundation artifacts are checkpoint directories written by `preference.fm::save_conjoint_foundation_bundle()`.

## Value

The bundle path (invisibly).

## See Also

[load\\_conjoint\\_foundation\\_bundle\(\)](#)

## Examples

```
## Not run:
preference.fm::save_conjoint_foundation_bundle(
  tempfile(),
  foundation_model = foundation_fit,
  overwrite = TRUE
)

## End(Not run)
```



---

save\_neural\_outcome\_bundle

*Save a portable neural outcome bundle*


---

## Description

Save a portable neural outcome bundle

## Usage

```
save_neural_outcome_bundle(
  file,
  theta_mean,
  theta_var = NULL,
  neural_model_info,
  names_list = NULL,
  p_list = NULL,
  factor_levels = NULL,
  mode = c("auto", "pairwise", "single"),
  fit_metrics = NULL,
  conda_env = "strategize_env",
  conda_env_required = TRUE,
  overwrite = FALSE,
  compress = TRUE,
  metadata = NULL
)
```

## Arguments

|                    |                                                                                |
|--------------------|--------------------------------------------------------------------------------|
| file               | Path to save the bundle (typically ending in .rds).                            |
| theta_mean         | Numeric vector of posterior means for neural parameters.                       |
| theta_var          | Optional numeric vector of posterior variances.                                |
| neural_model_info  | Neural model metadata (can include reticulate objects).                        |
| names_list         | Optional list of factor level names (see cs_prepare_W_encoding).               |
| p_list             | Optional p_list to derive factor level names when names_list is missing.       |
| factor_levels      | Optional integer vector of factor levels (derived from names_list by default). |
| mode               | "auto", "pairwise", or "single".                                               |
| fit_metrics        | Optional fit metrics to include.                                               |
| conda_env          | Conda env name for neural backend (stored in metadata).                        |
| conda_env_required | Require conda env to exist (stored in metadata).                               |
| overwrite          | Logical; overwrite existing file.                                              |
| compress           | Compression passed to saveRDS().                                               |
| metadata           | Optional list of extra metadata to include.                                    |

## Value

The bundle path (invisibly).

---

```
save_strategic_predictor
```

*Save a strategic predictor to disk*

---

### Description

Save a strategic predictor to disk

### Usage

```
save_strategic_predictor(
  fit,
  file,
  overwrite = FALSE,
  compress = TRUE,
  include_metrics = TRUE
)
```

### Arguments

|                              |                                                                  |
|------------------------------|------------------------------------------------------------------|
| <code>fit</code>             | A fitted <code>strategic_predictor</code> .                      |
| <code>file</code>            | Path to save the cache (typically ending in <code>.rds</code> ). |
| <code>overwrite</code>       | Logical; overwrite an existing file.                             |
| <code>compress</code>        | Compression passed to <code>saveRDS()</code> .                   |
| <code>include_metrics</code> | Logical; include out-of-sample fit metrics when present.         |

### Value

The cache path (invisibly).

---

```
strategic_prediction
```

*Fit a prediction-only outcome model*

---

### Description

Fit a prediction-only outcome model

### Usage

```
strategic_prediction(
  Y,
  W,
  X = NULL,
  ...,
  model = c("glm", "neural"),
  mode = c("auto", "pairwise", "single"),
  pair_id = NULL,
```

```

    profile_order = NULL,
    varcov_cluster_variable = NULL,
    conda_env = "strategize_env",
    conda_env_required = TRUE,
    neural_mcmc_control = NULL,
    use_regularization = TRUE,
    nFolds_glm = 3L,
    cache_path = NULL,
    cache_overwrite = FALSE,
    cache_compress = TRUE
  )

```

### Arguments

|                         |                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Y                       | Binary outcome in 0/1.                                                                                                                                                                                                                                                                                                                                             |
| W                       | Factor matrix/data.frame (one column per conjoint factor).                                                                                                                                                                                                                                                                                                         |
| X                       | Optional covariate matrix/data.frame (neural backend).                                                                                                                                                                                                                                                                                                             |
| ...                     | Reserved for future extensions.                                                                                                                                                                                                                                                                                                                                    |
| model                   | "glm" or "neural".                                                                                                                                                                                                                                                                                                                                                 |
| mode                    | "auto", "pairwise", or "single".                                                                                                                                                                                                                                                                                                                                   |
| pair_id                 | Optional pair identifier (required for pairwise).                                                                                                                                                                                                                                                                                                                  |
| profile_order           | Optional within-pair ordering (1/2).                                                                                                                                                                                                                                                                                                                               |
| varcov_cluster_variable | Optional cluster IDs for robust GLM vcov.                                                                                                                                                                                                                                                                                                                          |
| conda_env               | Conda env name for neural backend.                                                                                                                                                                                                                                                                                                                                 |
| conda_env_required      | Require conda env to exist (neural backend).                                                                                                                                                                                                                                                                                                                       |
| neural_mcmc_control     | Optional list passed to neural backend.                                                                                                                                                                                                                                                                                                                            |
| use_regularization      | Logical; run glinternet screening for GLM.                                                                                                                                                                                                                                                                                                                         |
| nFolds_glm              | Number of folds for glinternet CV (GLM).                                                                                                                                                                                                                                                                                                                           |
| cache_path              | Optional path to a cached predictor (.rds). If it exists and cache_overwrite is FALSE, the cached model is loaded instead of refitting. When a new model is fit, it is saved to this path. For neural fits, cache_path also enables restartable SVI checkpoints under paste0(cache_path, ".inprogress"); successful final saves remove that in-progress directory. |
| cache_overwrite         | Logical; refit and overwrite any existing cache at cache_path.                                                                                                                                                                                                                                                                                                     |
| cache_compress          | Compression setting passed to saveRDS().                                                                                                                                                                                                                                                                                                                           |

### Value

An object of class `strategic_predictor`.

---

|            |                                                                                            |
|------------|--------------------------------------------------------------------------------------------|
| strategize | <i>Estimate Optimal (or Adversarial) Stochastic Interventions for Conjoint Experiments</i> |
|------------|--------------------------------------------------------------------------------------------|

---

## Description

strategize implements the core methods described in the accompanying paper for learning an optimal or adversarial probability distribution over conjoint factor levels. It is specifically designed for forced-choice conjoint settings (e.g., candidate-choice experiments) and can accommodate scenarios in which a single agent optimizes its strategy in isolation, or in which two (potentially adversarial) agents simultaneously optimize against each other.

This function can be used to find the *optimal stochastic intervention* for maximizing an outcome of interest (e.g., vote choice, rating, or utility), possibly subject to a penalty that keeps the learned distribution close to the original design distribution. It can also incorporate institutional rules (e.g., primaries, multiple stages of choice) by specifying additional arguments. Estimation can be done under standard generalized linear modeling assumptions or more advanced approaches. The function returns estimates of the learned distribution and the associated performance quantity ( $Q(\pi^*)$ ) along with optional inference based on the (asymptotic) delta method.

## Usage

```
strategize(
  Y,
  W,
  X = NULL,
  lambda,
  varcov_cluster_variable = NULL,
  competing_group_variable_respondent = NULL,
  competing_group_variable_respondent_proportions = NULL,
  competing_group_variable_candidate = NULL,
  competing_group_competition_variable_candidate = NULL,
  pair_id = NULL,
  respondent_id = NULL,
  respondent_task_id = NULL,
  profile_order = NULL,
  p_list = NULL,
  slate_list = NULL,
  K = 1,
  nSGD = 100,
  diff = FALSE,
  adversarial = FALSE,
  adversarial_model_strategy = "four",
  include_stage_interactions = NULL,
  partial_pooling = NULL,
  partial_pooling_strength = 50,
  use_regularization = TRUE,
  force_gaussian = FALSE,
  force_reinforce = FALSE,
  a_init_sd = 0.001,
  outcome_model_type = "glm",
```

```

neural_mcmc_control = NULL,
penalty_type = "KL",
compute_se = FALSE,
se_method = c("full", "implicit"),
conda_env = "strategize_env",
conda_env_required = FALSE,
conf_level = 0.9,
nFolds_glm = 3L,
folds = NULL,
crossfit_q = FALSE,
crossfit_q_control = NULL,
nMonte_adversarial = 5L,
primary_pushforward = "mc",
primary_strength = 1,
primary_n_entrants = 1L,
primary_n_field = 1L,
nMonte_Qglm = 100L,
learning_rate_max = 0.001,
temperature = NULL,
save_outcome_model = FALSE,
presaved_outcome_model = FALSE,
outcome_model_key = NULL,
use_optax = FALSE,
optim_type = "gd",
optimism = "extragrad",
optimism_coef = 1,
rain_lambda = 1,
rain_gamma = 0.01,
rain_L = NULL,
rain_eta = 0.001,
rain_variant = "alg10_staged",
rain_output = "last",
compute_hessian = TRUE,
hessian_max_dim = 50L
)

```

## Arguments

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Y | A numeric or binary vector of observed outcomes, typically in $\{0, 1\}$ for forced-choice conjoint tasks, indicating whether the profile was selected. For instance, $Y = 1$ if candidate A was chosen over candidate B, and $Y = 0$ otherwise. The length must match the number of rows in W.                                                                                                                                          |
| W | A matrix or data frame representing the assigned levels of each factor in a conjoint design (one column per factor). Each row corresponds to a single profile. For forced-choice tasks, a given respondent may have contributed multiple rows if you reshape pairwise choices into long format. If the experiment used multiple factors $D$ , with each factor having $L_d$ levels, W should capture all factor assignments accordingly. |
| X | An optional matrix or data frame of additional covariates, often respondent-level features (e.g., respondent demographics). If $K > 1$ , X may be used internally to fit multi-cluster or multi-component outcome models, or to allow cluster-specific effect estimation for more granular insights. Defaults to NULL.                                                                                                                   |

|                                                              |                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>lambda</code>                                          | A numeric scalar or vector giving the regularization penalty (e.g., in Kullback-Leibler or L2 sense) used to shrink the learned probability distribution(s) of factor levels toward a baseline distribution <code>p_list</code> . Typically set via either domain knowledge or cross-validation.              |
| <code>varcov_cluster_variable</code>                         | An optional vector of cluster identifiers (e.g., respondent IDs) used to form a robust variance-covariance estimate of the outcome model. If NULL, the usual IID assumption is made. Defaults to NULL.                                                                                                        |
| <code>competing_group_variable_respondent</code>             | Optional variable marking competition group membership of respondents. Particularly relevant in adversarial settings ( <code>adversarial = TRUE</code> ) or multi-stage electoral settings, e.g., capturing the party of each respondent. Defaults to NULL.                                                   |
| <code>competing_group_variable_respondent_proportions</code> | Optional numeric vector specifying the population proportions of each competing group. If NULL, proportions are estimated from the data. Useful when the sample proportions differ from the target population proportions. Defaults to NULL.                                                                  |
| <code>competing_group_variable_candidate</code>              | Optional variable marking competition group membership of candidate profiles. Defaults to NULL.                                                                                                                                                                                                               |
| <code>competing_group_competition_variable_candidate</code>  | Optional variable indicating whether a candidate profile belongs to the competing group in adversarial settings. Defaults to NULL.                                                                                                                                                                            |
| <code>pair_id</code>                                         | A factor or numeric vector identifying the forced-choice pair. If each row of <code>W</code> is a single profile, <code>pair_id</code> groups the rows belonging to the same choice set. Defaults to NULL.                                                                                                    |
| <code>respondent_id, respondent_task_id</code>               | Another set of optional identifiers. <code>respondent_id</code> marks each respondent across tasks, while <code>respondent_task_id</code> can define unique IDs for repeated measurements from the same respondent across multiple tasks. Useful for advanced clustering or robust SEs. Defaults to NULL.     |
| <code>profile_order</code>                                   | If each forced-choice is shown with different ordering (e.g., Candidate A vs. Candidate B), <code>profile_order</code> can label each row accordingly. Helpful for ensuring consistent labeling of reference vs. opposing profiles. Defaults to NULL.                                                         |
| <code>p_list</code>                                          | An optional list describing the baseline probability distribution over factor levels in <code>W</code> . Typically derived from the initial design distribution or uniform assignment distribution. If NULL, the function may assume uniform or attempt to estimate the distribution from <code>W</code> .    |
| <code>slate_list</code>                                      | An optional list (or lists) providing custom slates of candidate features (and their associated probabilities). Used in more advanced or adversarial setups where certain combinations must be included or excluded. If NULL, no special constraints beyond the usual factor-level distributions are applied. |
| <code>K</code>                                               | Integer specifying the number of latent clusters for multi-component outcome models. If <code>K = 1</code> , no latent clustering is done. Defaults to 1.                                                                                                                                                     |
| <code>nSGD</code>                                            | Integer specifying the number of stochastic gradient descent (or gradient-based) iterations to use when learning the optimal distributions. Defaults to 100.                                                                                                                                                  |
| <code>diff</code>                                            | Logical indicating whether the outcome <code>Y</code> represents a first-difference or difference-based metric. In forced-choice contexts, typically <code>diff = FALSE</code> . Defaults to FALSE.                                                                                                           |

|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>adversarial</code>                | Logical controlling whether to enable the max-min adversarial scenario. When TRUE, the function searches for a pair of distributions (one for each competing party or group) such that each party's distribution is optimal given the other party's distribution. Defaults to FALSE.                                                                                                                                                                                                                      |
| <code>adversarial_model_strategy</code> | Character string indicating whether to estimate "four" outcome models (primary + general for each group), "two" outcome models (one per group reused for both primary and general), or "neural" (Bayesian Transformer models with party tokens; defaults to a single pooled model across groups and stages. Set <code>neural_mcmc_control\$n_bayesian_models = 2</code> to fit separate AST/DAG models). Defaults to "four".                                                                              |
| <code>include_stage_interactions</code> | Logical indicating whether to include stage (primary vs general) indicator and stage-by-factor interactions in the outcome model. When NULL (default), automatically set to TRUE when <code>adversarial_model_strategy = "two"</code> and FALSE otherwise. Including stage interactions allows a single pooled model to learn different response patterns for primary vs general election scenarios, which helps prevent pattern-matching equilibria where both parties converge to identical strategies. |
| <code>partial_pooling</code>            | Logical indicating whether to partially pool (shrink) group-specific outcome model coefficients toward a shared average when <code>adversarial_model_strategy = "two"</code> . When NULL (default), automatically set to TRUE for the two-strategy adversarial case and FALSE otherwise.                                                                                                                                                                                                                  |
| <code>partial_pooling_strength</code>   | Numeric scalar controlling the amount of shrinkage used for partial pooling in the two-strategy adversarial case. Interpreted as a pseudo-sample size: larger values increase pooling, smaller values preserve group differentiation. Defaults to 50.                                                                                                                                                                                                                                                     |
| <code>use_regularization</code>         | Logical indicating whether to regularize the outcome model (in addition to any penalty $\lambda$ on the distribution shift). This can help avoid overfitting in high-dimensional designs. Defaults to TRUE.                                                                                                                                                                                                                                                                                               |
| <code>force_gaussian</code>             | Logical indicating whether to force a Gaussian-based outcome modeling approach, even if Y is binary or forced-choice. If FALSE, the function attempts to choose a more appropriate link (e.g., "binomial"). Defaults to FALSE.                                                                                                                                                                                                                                                                            |
| <code>force_reinforce</code>            | Logical indicating whether to force REINFORCE-based optimization for the neural average-case objective when exact small-support enumeration is available. This only affects the optimization objective path; reported Q values continue to use the existing exact report-time evaluation when available. Defaults to FALSE.                                                                                                                                                                               |
| <code>a_init_sd</code>                  | Numeric scalar specifying the standard deviation for random initialization of unconstrained parameters used in the gradient-based search over factor-level probabilities. Defaults to 0.001.                                                                                                                                                                                                                                                                                                              |
| <code>outcome_model_type</code>         | Character string specifying the outcome model to use. Currently supports "glm" for generalized linear models or "neural" for a neural-network approximation. Defaults to "glm".                                                                                                                                                                                                                                                                                                                           |
| <code>neural_mcmc_control</code>        | Optional list overriding default MCMC settings used when <code>outcome_model_type = "neural"</code> . Named entries override the defaults in <code>two_step_model_outcome_neural.R</code> .                                                                                                                                                                                                                                                                                                               |

Set `neural_mcmc_control$uncertainty_scope = "output"` to compute delta-method uncertainty using only the output-layer parameters (default is "all"). In adversarial neural mode, set `neural_mcmc_control$n_bayesian_models = 2` to fit separate AST/DAG models (default is 1 for a single differential model). Use `neural_mcmc_control$ModelDims` and `neural_mcmc_control$ModelDepth` to override the Transformer hidden width and depth. Pairwise neural fits default to `neural_mcmc_control$cross_candidate_encoder = "term"` when unspecified, except rank-positive low-rank respondent-candidate interaction defaults to "none" to avoid duplicating pairwise interaction capacity. Set `neural_mcmc_control$cross_candidate_encoder = "term"` (or TRUE) to include the opponent-dependent cross-candidate term in pairwise mode, set `neural_mcmc_control$cross_candidate_encoder = "attn"` to add a lightweight cross-attention step. When combined with `neural_mcmc_control$residual_mode = "full_attn"`, that step consumes the depth-attended candidate-token readout. Set `neural_mcmc_control$cross_candidate_encoder = "full"` to enable a full cross-encoder that jointly encodes both candidates. Use "none" (or FALSE) to disable. Set `neural_mcmc_control$qk_norm = TRUE` (default) to apply RMSNorm to projected queries and keys before forming attention logits. Set `neural_mcmc_control$residual_mode = "full_attn"` to replace the default additive/ReZero residual path with full depth-wise attention over all prior layer outputs plus a final depth-attentive readout; use "standard" (default) to keep the existing residual formulation. Rank-positive low-rank Bernoulli pairwise logits use RMS logit normalization by default; set `neural_mcmc_control$low_rank_logit_normalization = "none"` to disable it. Explicit `low_rank_logit_transform = "softclip"` remains supported for compatibility. For variational inference (`subsample_method = "batch_vi"`), set `neural_mcmc_control$optimizer` to "muon" (default when `optax.contrib.muon` is available), "adam" (`numpyro.optim`), "adamw" (AdamW), or "adabelief" (`optax`). Muon is only applied to matrix-valued model-weight locations when the guide preserves matrix structure; with `vi_guide = "auto_diagonal"`, it falls back to AdamW/Adam. Learning-rate decay is controlled by `neural_mcmc_control$svi_steps` (integer steps, or "optimal" for a scaling-law heuristic based on model/data size; for minibatched VI this also scales with `batch_size`) and `neural_mcmc_control$svi_lr_scheduler` (default "warmup\_cosine"), with optional `svi_lr_warmup_frac` and `svi_lr_end_factor`. Validation-based early stopping is enabled by default for SVI-backed neural fits; set `neural_mcmc_control$early_stopping = FALSE` to force the full `svi_steps` budget. Use `neural_mcmc_control$early_stopping_n_checks` (default 10) to request an approximate number of validation checks during SVI early stopping; the cadence is derived as `ceiling(svi_steps / early_stopping_n_checks)`. Use `neural_mcmc_control$early_stopping_patience` (default 3) to control how many consecutive non-improving validation checks are tolerated before stopping. For compact streaming SVI, `early_stopping = TRUE` enables validation best-checkpoint selection but still runs the full `svi_steps` budget. Set `neural_mcmc_control$early_stopping_validation_frac` (default 0.05) to retain approximately that fraction of evaluable observations in the held-out validation split, and optionally `neural_mcmc_control$early_stopping_validation_max_n` (default 2048) to cap the retained validation size after fraction-based sizing. Set `early_stopping_validation_max_n = NULL` to disable that cap. Checkpoint gradient diagnostics are enabled by default for SVI fits and are evaluated at the early-stopping checkpoint cadence; set `neural_mcmc_control$gradient_diagnostics = FALSE` to disable them. Advanced restart checkpoints can be enabled with `neural_mcmc_control$checkpoint_path`; rerunning with the same data and controls resumes from `latest.rds` and promotes the best validation checkpoint when present. `checkpoint_resume` defaults to TRUE when a checkpoint path



|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | is present, <code>checkpoint_n_checks</code> defaults to 10 for non-early-stopping checkpoint cadence, and <code>checkpoint_compress</code> defaults to FALSE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <code>penalty_type</code>        | A character string specifying the type of penalty (e.g., "KL", "L2", or "LogMaxProb") used in the objective function for shifting the factor-level probabilities away from the baseline <code>p_list</code> . Defaults to "KL".                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>compute_se</code>          | Logical indicating whether standard errors should be computed for the final estimates (via the delta method or related expansions). Defaults to FALSE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <code>se_method</code>           | Character string specifying the SE computation method when <code>compute_se = TRUE</code> . "full" differentiates through the full optimization trace (default). "implicit" uses implicit differentiation at the solution (adversarial equilibrium or non-adversarial optimum).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>conda_env</code>           | A character string naming a Python conda environment that includes <b>jax</b> , <b>optax</b> , and other dependencies used by the optimization workflow. If not NULL, the function attempts to activate that environment. Defaults to "strategize_env".                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <code>conda_env_required</code>  | Logical; if TRUE, raises an error if the environment given by <code>conda_env</code> cannot be activated. Otherwise, the function attempts to proceed with any available installation. Defaults to FALSE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <code>conf_level</code>          | Numeric in (0, 1), specifying the confidence level for intervals or credible bounds. Defaults to 0.90.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <code>nFolds_glm</code>          | Integer specifying the number of folds (default 3L) for internal cross-validation used in certain outcome model or regularization steps. Defaults to 3L.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <code>folds</code>               | An optional user-supplied partitioning or CV scheme, overriding <code>nFolds_glm</code> . Defaults to NULL.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <code>crossfit_q</code>          | Logical. If TRUE, compute a cross-fitted policy-value diagnostic for supported pairwise forced-choice binomial GLM runs. This currently includes non-adversarial average-case runs and adversarial <code>adversarial_model_strategy = "four"</code> runs with exactly two respondent and candidate groups. The returned fields <code>Q_crossfit</code> , <code>Q_reference_crossfit</code> , and <code>Q_gain_crossfit</code> summarize out-of-fold value for the learned policy relative to the assignment-policy baseline.                                                                                                                                                                                                                                                                  |
| <code>crossfit_q_control</code>  | Optional list controlling cross-fitted Q evaluation. Supported entries include <code>folds</code> , <code>seed</code> , <code>split_by</code> , <code>estimators</code> , <code>headline</code> , <code>weight_clip</code> , <code>n_policy_draws</code> , <code>chunk_size</code> , <code>return_fold_results</code> , and <code>perspective_group</code> . For adversarial runs, <code>perspective_group</code> selects the candidate group whose win probability is reported; if omitted, the first sorted group label is used. The adversarial estimator assumes independent group-specific profile assignment under <code>p_list</code> . The default headline estimator is doubly robust ("dr"), with IPS, SNIPS, and model-only diagnostics returned in <code>Q_crossfit_info</code> . |
| <code>nMonte_adversarial</code>  | Integer specifying the number of Monte Carlo samples used in adversarial or max-min steps, e.g., sampling from the opposing candidate's distribution to approximate expected payoffs. Defaults to 5L.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <code>primary_pushforward</code> | Character string controlling the primary-stage push-forward estimator. Use "mc" (default) for Monte Carlo sampling with per-draw primary winners, or "linearized" for the faster averaged-weight approximation, or "multi" for multi-candidate primaries.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| primary_strength       | Numeric scalar controlling primary decisiveness. Values $> 1$ make primary outcomes more deterministic; values in $(0, 1)$ make primaries more noisy. Defaults to 1.0 (neutral scaling).                                                                                                                                                                                                                                                                                                           |
| primary_n_entrants     | Integer number of entrant candidates sampled per party in multi-candidate primaries ( <code>primary_pushforward = "multi"</code> ). Defaults to 1.                                                                                                                                                                                                                                                                                                                                                 |
| primary_n_field        | Integer number of field candidates sampled per party in multi-candidate primaries ( <code>primary_pushforward = "multi"</code> ). Defaults to 1.                                                                                                                                                                                                                                                                                                                                                   |
| nMonte_Qglm            | Integer specifying the number of Monte Carlo samples for evaluating or approximating the quantity of interest under certain outcomes or distributions. Defaults to 100L.                                                                                                                                                                                                                                                                                                                           |
| learning_rate_max      | Base learning rate for gradient-based optimizers. Defaults to 0.001.                                                                                                                                                                                                                                                                                                                                                                                                                               |
| temperature            | Numeric temperature parameter used in Gumbel-Softmax sampling to smooth the exploration of the probability simplex. Smaller values yield distributions closer to the argmax. Defaults to 0.5.                                                                                                                                                                                                                                                                                                      |
| save_outcome_model     | Logical indicating whether to save the fitted outcome model to disk for reuse. Useful for large models or repeated runs. Defaults to FALSE.                                                                                                                                                                                                                                                                                                                                                        |
| presaved_outcome_model | Logical indicating whether to use a previously saved outcome model instead of re-fitting. Defaults to FALSE.                                                                                                                                                                                                                                                                                                                                                                                       |
| outcome_model_key      | Optional character string to append to saved outcome model filenames. Useful for distinguishing between different model configurations or experimental runs. When provided, files are saved as <code>main_{group}_{round}_{key}.csv</code> . Defaults to NULL.                                                                                                                                                                                                                                     |
| use_optax              | Logical indicating whether to use the <code>optax</code> library for gradient-based optimization in JAX (TRUE) or a built-in method (FALSE). Defaults to FALSE.                                                                                                                                                                                                                                                                                                                                    |
| optim_type             | A character string for choosing which optimizer or approach is used internally (e.g., "gd" for gradient descent). Defaults to "gd".                                                                                                                                                                                                                                                                                                                                                                |
| optimism               | Character string controlling optimistic / extra-gradient updates for the gradient optimizer. Options: "extragrad" (default; classic Korpelevich extra-gradient), "smp" (stochastic mirror-prox: extra-gradient with weighted averaging of look-ahead points), "ogda" (optimistic gradient), "rain" (RAIN: Recursive Anchored Iteration with anchored extra-gradient and increasing quadratic anchor penalties), or "none" (standard updates). Only supported when <code>use_optax = FALSE</code> . |
| optimism_coef          | Numeric scalar controlling the magnitude of optimism adjustments. For "ogda", this scales the optimistic correction term. Ignored when <code>optimism = "rain"</code> .                                                                                                                                                                                                                                                                                                                            |
| rain_lambda            | Numeric scalar giving the base RAIN regularization scale $\lambda$ . The staged algorithm uses $\lambda_0 = \gamma\lambda$ and grows by $(1+\gamma)$ each stage. Only used when <code>optimism = "rain"</code> .                                                                                                                                                                                                                                                                                   |
| rain_gamma             | Non-negative numeric scalar for the RAIN anchor-growth parameter $\gamma$ . Larger values grow anchor weights faster. If not supplied, the default is auto-scaled downward when nSGD exceeds 100 to keep total anchor growth roughly constant. Only used when <code>optimism = "rain"</code> .                                                                                                                                                                                                     |

|                 |                                                                                                                                                                                                                                                                                                    |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rain_L          | Optional numeric scalar for the Lipschitz constant $L$ used by RAIN. When supplied and rain_eta is missing, rain_eta defaults to $1/(8L)$ and stage lengths follow $T_s = 16L/\lambda_s$ . If NULL, $L$ is estimated from rain_eta. Only used when optimism = "rain".                              |
| rain_eta        | Optional numeric scalar step size $\eta$ for RAIN. Defaults to 0.001 and is auto-scaled downward when nSGD exceeds 100 if not supplied. If rain_L is supplied and rain_eta is missing, defaults to $1/(8L)$ . Only used when optimism = "rain".                                                    |
| rain_variant    | Character string specifying the RAIN variant. Options: "alg10_staged" (merged Algorithm 10 with recursive stage anchors; default) or "alg9_single_loop" (single-loop variant; not yet implemented). Only used when optimism = "rain".                                                              |
| rain_output     | Character string controlling the stage output for RAIN. "uniform_half" samples uniformly from half-iterates within the stage (most faithful); "last" returns the last iterate (default). Only used when optimism = "rain".                                                                         |
| compute_hessian | Logical. Whether to compute Hessian functions for equilibrium geometry analysis in adversarial mode. When TRUE (default), Hessian functions are JIT-compiled to enable <a href="#">check_hessian_geometry</a> analysis. Set to FALSE to skip Hessian computation for faster execution.             |
| hessian_max_dim | Integer. Maximum number of parameters per player before automatically skipping Hessian computation. Defaults to 50L. For problems with more parameters, Hessian computation is skipped to avoid memory/time overhead. The result will have hessian_skipped_reason = "high_dimension" in this case. |

## Details

**Modeling the outcome:** Internally, strategize may fit a generalized linear model or a more flexible approach (such as multi-cluster factorization) to learn the mapping from factor-level assignments  $W$  (and optional covariates  $X$ ) onto outcomes  $Y$ . Once these outcome coefficients are estimated, the function uses gradient-based or closed-form solutions to find the *optimal stochastic intervention*(s), i.e., new factor-level probability distributions that maximize an expected outcome (or solve the max-min adversarial problem).

**Adversarial or strategic design:** When adversarial = TRUE, the function attempts to solve a zero-sum game in which one agent (say, "A") chooses its distribution to maximize vote share, while the other ("B") simultaneously chooses its distribution to minimize "A"'s vote share. In many settings, competing\_group\_variable\_respondent and related arguments help define which respondents belong to the "A" or "B" sub-electorate (e.g., a primary). The final solution is a mixed-strategy Nash equilibrium, if it exists, for the forced-choice environment. This can be used to compare or interpret real-world candidate positioning in multi-stage elections.

**Regularization:** The argument lambda penalizes how far the learned distribution strays from the baseline distribution p\_list. This helps avoid overfitting in high-dimensional designs. Different penalty types can be selected via penalty\_type.

**Implementation details:** Under the hood, this function may rely on **jax** for automatic differentiation. By default, it uses an internal gradient-based approach. If use\_optax = TRUE, the optax library is used for optimization. The function can automatically detect or load a **conda** environment if specified, though advanced users can pass conda\_env\_required = TRUE to enforce that environment activation is mandatory.

**Value**

A named list containing:

`pi_star_point` An estimate of the (possibly multi-cluster or adversarial) optimal distribution(s) over the factor levels.

Structure depends on parameters:

- If `adversarial = TRUE` and `K = 1`, returns a pair of distributions (e.g., maximin solutions).
- If `K > 1`, returns a list where each element corresponds to a cluster-optimal distribution.
- Otherwise, returns a single distribution.

`pi_star_se` Standard errors for entries in `pi_star_point`. Mirrors the structure of `pi_star_point` (e.g., a pair of SEs if `adversarial = TRUE` and `K = 1`). Only present if `compute_se = TRUE`.

`Q_point` Primary point estimate(s) of the optimized outcome (e.g., utility or vote share). Matches the structure of `pi_star_point`.

`Q_se` Primary standard errors for `Q_point`. Only present if `compute_se = TRUE`.

`Q_point_mEst`, `Q_se_mEst` Backward-compatible aliases for `Q_point` and `Q_se`.

`pi_star_lb`, `pi_star_ub` Confidence bounds for `pi_star_point` (if `compute_se = TRUE` and a confidence level is provided).

`outcome_model_view` Interpretable summaries of the fitted outcome models (by player and stage). Includes main-effect tables and top interactions for AST/DAG primary/general submodels when available.

`CVInfo` Cross-validation performance data (if applicable). Typically a `data.frame` or list.

`Q_crossfit`, `Q_reference_crossfit`, `Q_gain_crossfit` Optional cross-fitted policy-value diagnostics returned when `crossfit_q = TRUE`.

`Q_crossfit_info` Optional fold-level and estimator-level diagnostics for cross-fitted Q evaluation, including IPS/SNIPS/model/DR summaries and importance-weight effective sample size. For adversarial runs this also records the perspective/opponent groups, target and reference summaries, and the independent-product assignment assumption used for profile weights.

`estimationType` String indicating the approach used by the active estimator, "TwoStep".

... Additional internal details (e.g., fitted models, optimization logs).

**See Also**

[cv\\_strategize](#) for cross-validation across candidate values of `lambda`.

**Examples**

```
# =====
# Example 1: Basic single-agent optimization
# =====
# Generate synthetic conjoint data
set.seed(42)
n <- 400 # Number of profiles (200 pairs)

# Factor matrix: candidate attributes
W <- data.frame(
  Gender = sample(c("Male", "Female"), n, replace = TRUE),
  Age = sample(c("Young", "Middle", "Old"), n, replace = TRUE),
  Party = sample(c("Dem", "Rep"), n, replace = TRUE)
)
```

```

# Simulate outcome: Female + Young candidates preferred
latent <- 0.3 * (W$Gender == "Female") +
  0.2 * (W$Age == "Young") -
  0.1 * (W$Age == "Old")
prob <- plogis(latent)

# Paired forced-choice: within each pair, one wins
pair_id <- rep(1:(n/2), each = 2)
Y <- numeric(n)
for (p in unique(pair_id)) {
  idx <- which(pair_id == p)
  winner <- sample(idx, 1, prob = prob[idx])
  Y[idx] <- as.integer(seq_along(idx) == which(idx == winner))
}
profile_order <- rep(1:2, n/2)

# Baseline probabilities (uniform assignment)
p_list <- list(
  Gender = c(Male = 0.5, Female = 0.5),
  Age = c(Young = 1/3, Middle = 1/3, Old = 1/3),
  Party = c(Dem = 0.5, Rep = 0.5)
)

# Run strategize to find optimal distribution
# (requires conda environment with JAX - see build_backend())
result <- strategize(
  Y = Y,
  W = W,
  lambda = 0.1,
  pair_id = pair_id,
  respondent_id = pair_id,
  respondent_task_id = pair_id,
  profile_order = profile_order,
  p_list = p_list,
  diff = TRUE,
  nSGD = 50,
  compute_se = FALSE
)

# View optimal distribution
print(result$pi_star_point)

# View expected outcome under optimal strategy
print(result$Q_point)

```

---

strategize.plot

---

*Plot Estimated Probabilities for Hypothetical Scenarios*


---

## Description

This function creates a grid of base R plots to visualize and compare probabilities (and optionally their confidence intervals) across multiple hypothetical or assignment scenarios. By default, it arranges the plots in a 3xN grid, labeling each row according to the factor or condition being displayed.

**Usage**

```

strategize.plot(
  pi_star_list = NULL,
  pi_star_se_list = NULL,
  p_list = NULL,
  col.main = "black",
  cex.main = 1.5,
  zStar = 1,
  xlim = NULL,
  ticks_type = "assignmentProbs",
  col_vec = NULL,
  plot_names = TRUE,
  plot_ci = TRUE,
  widths_vec,
  heights_vec,
  main_title = "",
  margins_vec = NULL,
  add = FALSE,
  pch = 20,
  factor_name_transformer = function(x) {
    x
  },
  level_name_transformer = function(x) {
    x
  },
  label_wrap_width = 30,
  label_line_height = 0.6,
  cex.axis = 1.25,
  open_browser = FALSE
)

```

**Arguments**

|                              |                                                                                                                                                                                                                             |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>pi_star_list</code>    | A list of numeric vectors, each corresponding to a set of hypothetical probabilities to be plotted. These are typically model-based or derived values.                                                                      |
| <code>pi_star_se_list</code> | A list of numeric vectors of the same structure as <code>pi_star_list</code> , containing standard errors for each probability. Used to plot confidence intervals.                                                          |
| <code>p_list</code>          | A list of numeric vectors of "assignment" or baseline probabilities to be overlaid as vertical ticks on each plot (depending on <code>ticks_type</code> ).                                                                  |
| <code>col.main</code>        | Character. Color for the main title in each subplot. Default is "black".                                                                                                                                                    |
| <code>cex.main</code>        | Numeric. Character expansion factor for main titles. Default is 1.5.                                                                                                                                                        |
| <code>zStar</code>           | Numeric. Multiplier for the standard error bars (e.g., 1.96 for approximately 95 percent confidence intervals). Default is 1.                                                                                               |
| <code>xlim</code>            | Numeric vector of length 2. The x-axis limits for all subplots. Defaults to <code>c(0, 1)</code> if not specified.                                                                                                          |
| <code>ticks_type</code>      | Character. Controls the type of reference ticks added. "assignmentProbs": Vertical ticks drawn at the positions from <code>p_list</code> (default). "zero": Vertical ticks drawn at 0. "none": No vertical reference ticks. |

|                         |                                                                                                                                             |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| col_vec                 | Optional character vector of colors (one per set of probabilities in pi_star_list). If NULL, uses sequential indexing for color.            |
| plot_names              | Logical. If TRUE (default), factor/condition labels will be placed along the y-axis.                                                        |
| plot_ci                 | Logical. If TRUE (default), error bars will be drawn using pi_star_se_list.                                                                 |
| widths_vec, heights_vec | Currently unused. Reserved for future layout expansions.                                                                                    |
| main_title              | Character. An overall title for the plot. Default is an empty string.                                                                       |
| margins_vec             | Currently unused. Reserved for future layout expansions.                                                                                    |
| add                     | Logical. If FALSE (default), a new plot is created. If TRUE, points/error bars are added to an existing plot space.                         |
| pch                     | Numeric or character. The plotting symbol. Default is 20.                                                                                   |
| factor_name_transformer | Function to transform factor names for display. Should accept and return a character vector. Default is identity function.                  |
| level_name_transformer  | Function to transform level names for display. Should accept and return a character vector. Default is identity function.                   |
| label_wrap_width        | Numeric. If provided, long level labels are wrapped to this character width using line breaks. Default is 30. Use NULL to disable wrapping. |
| label_line_height       | Numeric. Additional vertical spacing (in plot units) added per wrapped line beyond the first. Default is 0.6.                               |
| cex.axis                | Numeric. Character expansion factor for axis labels. Default is 1.25.                                                                       |
| open_browser            | Logical. If TRUE, opens a browser for debugging. Default is FALSE. Intended for development use only.                                       |

## Details

strategize.plot arranges multiple subplots (3 columns by default) in a grid that depends on the number of elements in p\_list. Each subplot will show a factor level or condition on the y-axis, with probabilities along the x-axis. If confidence intervals are provided (pi\_star\_se\_list), horizontal error bars around each probability point will be displayed. Additionally, vertical reference ticks can be added, showing values from p\_list or zero depending on ticks\_type.

## Value

Invisibly returns NULL. This function is primarily called for its side effect: producing a multi-panel base R plot.

## Examples

```
# =====
# Visualize optimal vs baseline distributions
# =====
# This function works without JAX - just needs the result structure

# Create mock strategize result for plotting
# (In practice, use output from strategize())
```

```

pi_star_list <- list(k1 = list(
  Gender = c(Male = 0.35, Female = 0.65),
  Age = c(Young = 0.45, Middle = 0.30, Old = 0.25),
  Party = c(Dem = 0.40, Rep = 0.60)
))

pi_star_se_list <- list(k1 = list(
  Gender = c(Male = 0.04, Female = 0.04),
  Age = c(Young = 0.03, Middle = 0.03, Old = 0.03),
  Party = c(Dem = 0.05, Rep = 0.05)
))

# Baseline (original assignment) probabilities
p_list <- list(
  Gender = c(Male = 0.5, Female = 0.5),
  Age = c(Young = 0.33, Middle = 0.33, Old = 0.34),
  Party = c(Dem = 0.5, Rep = 0.5)
)

# Plot comparing optimal to baseline
strategize.plot(
  pi_star_list = pi_star_list,
  pi_star_se_list = pi_star_se_list,
  p_list = p_list,
  main_title = "Optimal vs Baseline Distribution",
  ticks_type = "assignmentProbs" # Show baseline as reference ticks
)

```

---

strategize\_fm\_backend *Backend bridge for preference foundation-model training*

---

## Description

Returns the small set of internal neural/JAX helpers needed by external foundation-model training packages. This is a developer bridge for **preference.fm**; user workflows should call the public strategize APIs.

## Usage

```
strategize_fm_backend()
```

## Value

A named list of helper functions and the shared JAX environment.



---

|                   |                                           |
|-------------------|-------------------------------------------|
| strategize_preset | <i>Get Recommended Parameter Settings</i> |
|-------------------|-------------------------------------------|

---

**Description**

Returns a list of parameter values suitable for passing to `strategize()`. This simplifies the API by providing sensible defaults for common use cases.

**Usage**

```
strategize_preset(
  preset = c("standard", "quick_test", "publication", "adversarial")
)
```

**Arguments**

|        |                                                                                                                                                                                                                                                                                    |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| preset | One of:<br>"quick_test" Minimal iterations for testing code (not for inference)<br>"standard" Reasonable defaults for most analyses (default)<br>"publication" Higher accuracy for publication-quality results<br>"adversarial" Settings tuned for adversarial/game-theoretic mode |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Details**

These presets provide starting points. You should still set data-specific parameters like  $\gamma$ ,  $W$ , `p_list`, and potentially `lambda`.

For the "publication" preset, `lambda` is not included because you should use `cv_strategize()` to select it via cross-validation.

**Value**

Named list of parameters that can be merged with `strategize()` arguments.

**Examples**

```
# Get standard settings
params <- strategize_preset("standard")
print(params)

# Use with strategize (hypothetically)
# result <- do.call(strategize, c(list( $\gamma$  =  $\gamma$ ,  $W$  =  $W$ , p_list = p_list), params))
```

---

summarize\_adversarial *Summarize Adversarial Strategize Results*


---

## Description

Prints a comprehensive summary of adversarial equilibrium results, including strategies, equilibrium quality metrics, and four-quadrant breakdown.

## Usage

```
summarize_adversarial(
  result,
  validate = TRUE,
  check_hessian = TRUE,
  verbose = TRUE
)
```

## Arguments

|               |                                                                                                                                     |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------|
| result        | Output from <code>strategize</code> with <code>adversarial = TRUE</code>                                                            |
| validate      | Logical. Whether to run equilibrium validation. Default is TRUE. Set to FALSE for faster output without validation.                 |
| check_hessian | Logical. Whether to include Hessian geometry analysis. Default is TRUE. Only runs if Hessian functions are available in the result. |
| verbose       | Logical. Whether to print the summary. Default is TRUE.                                                                             |

## Details

This function provides a unified view of adversarial equilibrium results:

- Equilibrium vote share  $Q^*$  with standard error
- Optimized strategies for both parties (AST and DAG)
- Equilibrium quality metrics (best-response errors)
- Four-quadrant contribution breakdown
- Voter proportion information
- Convergence status

## Value

Invisibly returns a list containing all summary statistics.

## Examples

```
## Not run:
# Run adversarial strategize
result <- strategize(Y = y, W = w, adversarial = TRUE, nSGD = 500)

# Print summary
summarize_adversarial(result)
```

```
# Quick summary without validation
summarize_adversarial(result, validate = FALSE)

## End(Not run)
```

---

```
summary.strategize_result
```

*Summary Method for strategize Results*

---

### Description

Creates a summary table comparing baseline and optimal distributions.

### Usage

```
## S3 method for class 'strategize_result'
summary(object, ...)
```

### Arguments

|        |                                |
|--------|--------------------------------|
| object | A strategize result object     |
| ...    | Additional arguments (ignored) |

### Value

Invisibly returns a data.frame with the comparison

---

```
validate_equilibrium
```

*Validate Nash Equilibrium Quality*

---

### Description

Computes best-response error to verify that the optimized strategies form a Nash equilibrium. At a true Nash equilibrium, neither player can improve their payoff by unilaterally changing strategy.

### Usage

```
validate_equilibrium(
  result,
  method = c("grid", "gradient"),
  resolution = 50,
  tolerance = 0.01,
  nMonte = 100,
  plot = TRUE,
  verbose = TRUE,
  seed = 1L
)
```

## Arguments

|            |                                                                                                                                                                                                                                                                                                                       |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| result     | Output from <code>strategize</code> with <code>adversarial = TRUE</code>                                                                                                                                                                                                                                              |
| method     | Character string specifying the search method: <ul style="list-style-type: none"> <li>• "grid": Grid search around the current solution (deterministic for 1-2 parameters; random sampling for higher dimensions)</li> <li>• "gradient": Gradient ascent from current solution (faster)</li> </ul> Default is "grid". |
| resolution | Integer. Number of grid points per dimension for grid search. Default is 50.                                                                                                                                                                                                                                          |
| tolerance  | Numeric. Maximum BR error to consider as equilibrium. Default is 0.01 (i.e., neither player can improve vote share by more than 1%).                                                                                                                                                                                  |
| nMonte     | Integer. Number of Monte Carlo samples for Q evaluation. This overrides the Monte Carlo settings stored in <code>result</code> for the duration of this validation call. Default is 100.                                                                                                                              |
| plot       | Logical. Whether to generate visualization. Default is TRUE.                                                                                                                                                                                                                                                          |
| verbose    | Logical. Whether to print progress messages. Default is TRUE.                                                                                                                                                                                                                                                         |
| seed       | Integer seed for reproducible Monte Carlo evaluation and deterministic search behavior. Use NULL for non-deterministic behavior. Default is 1.                                                                                                                                                                        |

## Details

### What is a Nash Equilibrium?

In the adversarial setting, two parties (AST and DAG) simultaneously choose probability distributions over candidate attributes (e.g., gender, age, policy positions). Each party wants to maximize their expected vote share given the opponent's strategy. A Nash equilibrium is a pair of strategies where:

- AST's strategy is optimal given DAG's strategy
- DAG's strategy is optimal given AST's strategy

At equilibrium, neither party can improve by unilaterally changing their strategy.

### What is Best-Response Error?

The best-response error measures how far a player's current strategy is from being optimal. For player  $p$ , it is defined as:

$$BR\_error_p = \max_{\pi_p} Q(\pi_p, \pi_{-p}^*) - Q(\pi_p^*, \pi_{-p}^*)$$

In words: if we fix the opponent's strategy and search for the best possible response, how much better could we do compared to our current strategy?

- **BR error = 0:** The player is already playing optimally (true equilibrium)
- **BR error > 0:** The player could improve by switching strategies
- **BR error = 0.05:** The player could gain 5 percentage points in vote share

### How Validation Works

This function performs the following steps:

1. Evaluates  $Q$  (expected vote share) at the current solution
2. For AST: fixes DAG's strategy and searches for AST's best response

3. For DAG: fixes AST's strategy and searches for DAG's best response
4. Computes the improvement each player could achieve (BR error)
5. If both BR errors are below tolerance, declares it a valid equilibrium

### Interpretation

- `is_equilibrium = TRUE`: The solution is a valid Nash equilibrium (within numerical tolerance). Both parties are playing optimally.
- `is_equilibrium = FALSE`: At least one party could improve by changing strategy. This may indicate insufficient SGD iterations, a local minimum, or numerical issues.

### Search Methods

- `"grid"`: Searches over a discretized grid of strategies around the current solution. More thorough but slower. Recommended for validation.
- `"gradient"`: Runs additional gradient ascent steps from the current solution. Faster but may miss improvements in other directions.

### Value

A list containing:

**br\_error\_ast** Best-response error for AST player (how much AST could improve by switching to best response)

**br\_error\_dag** Best-response error for DAG player

**is\_equilibrium** Logical. TRUE if both errors are below tolerance

**Q\_current** Current objective value at the solution

**Q\_br\_ast** Objective value if AST switched to best response

**Q\_br\_dag** Objective value if DAG switched to best response

**br\_strategy\_ast** The best response strategy for AST (for comparison)

**br\_strategy\_dag** The best response strategy for DAG

**nMonte** Monte Carlo sample count used for validation

**seed** Seed used for deterministic evaluation (if provided)

**plot** ggplot object if `plot=TRUE` and `ggplot2` available, else `NULL`

### Examples

```
## Not run:
# Run adversarial strategize
result <- strategize(Y = y, W = w, adversarial = TRUE, nSGD = 500)

# Validate equilibrium
validation <- validate_equilibrium(result)
print(validation$is_equilibrium)
print(validation$br_error_ast)
print(validation$br_error_dag)

# If validation fails, try more SGD iterations
if (!validation$is_equilibrium) {
  result2 <- strategize(Y = y, W = w, adversarial = TRUE, nSGD = 2000)
  validation2 <- validate_equilibrium(result2)
```

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*validate\_equilibrium*

}

## End(Not run)

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